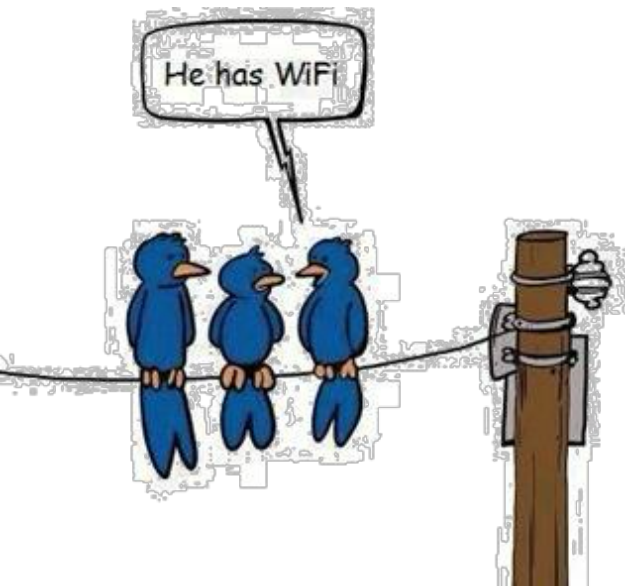


ECSE 4560/ 6560

Modern Communication Systems

Lecture 15: Midterm Review and MIMO Intro

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Midterm Review**
- Projects
 - Group formation [MCS Fall25 project.xlsx](#)
- HW/Lab 4 due Wed ~~Oct 22~~ (Late due Friday Oct 24). *Monday 24th*
- Exam 1: Friday Oct 24 -- practice problems coming soon *HS → Friday*
- Exam 2: Friday Nov 21
- Project mid-term report due Friday Nov 7
- Final Project Presentation and Demos: Dec 9 and Dec 12
- Project Report and peer evaluation: Finals exam week

Extra Reading on Piazza

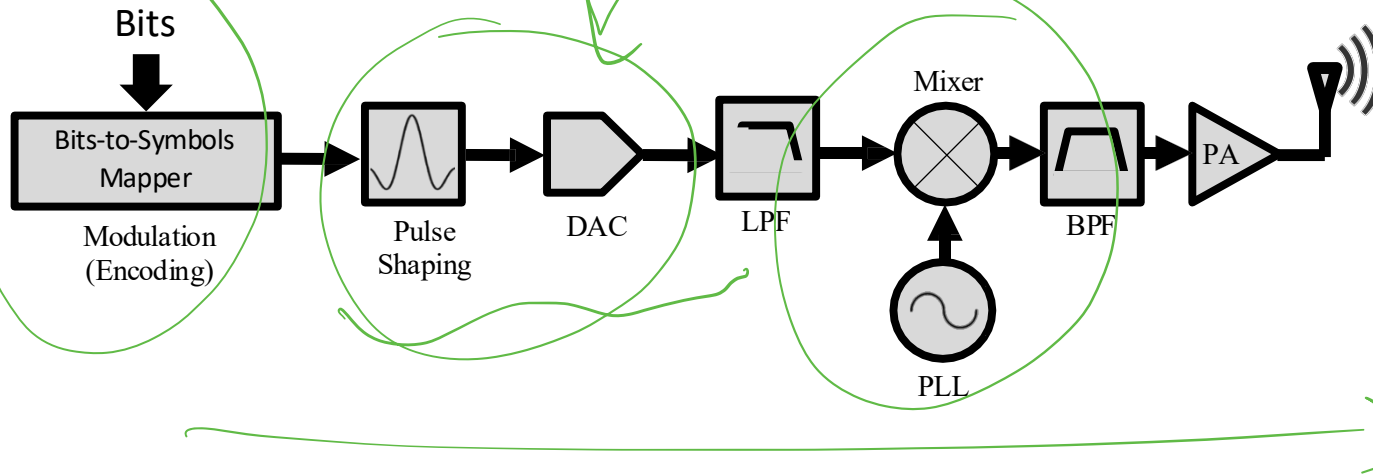
- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - MIMO Chapter 6
- An Adaptive Orthogonal Frequency Division Multiplexing Baseband Modem for Wideband Wireless Channels; by Jit Ken Tan, MIT Thesis
 - OFDM Decoding and Synchronization
 - Lab 4 Implementation

Today's Class

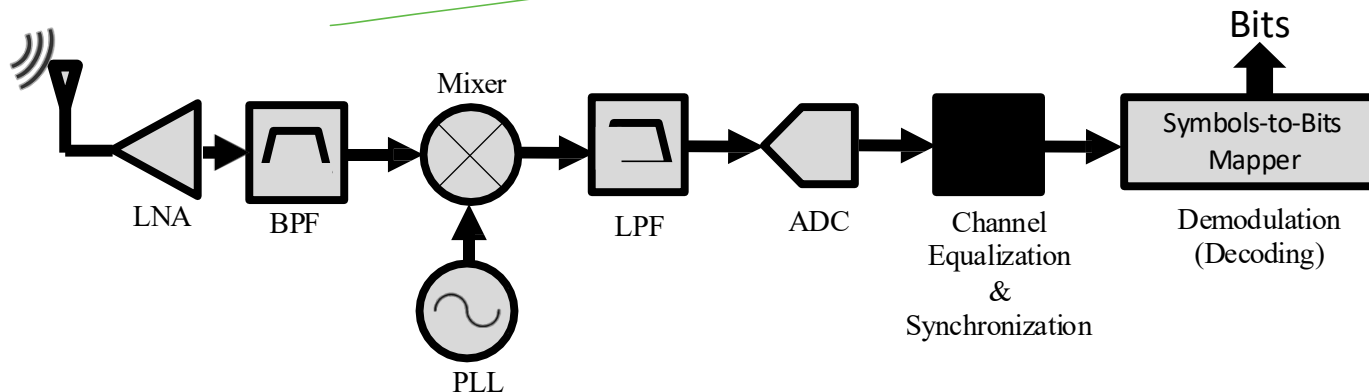
- Midterm Review
 - Sample Midterm 1 Questions
- MIMO Intro
 - MIMO in WiFi Standards
 - MIMO Intro
 - MIMO vector representation

Wireless Communication System

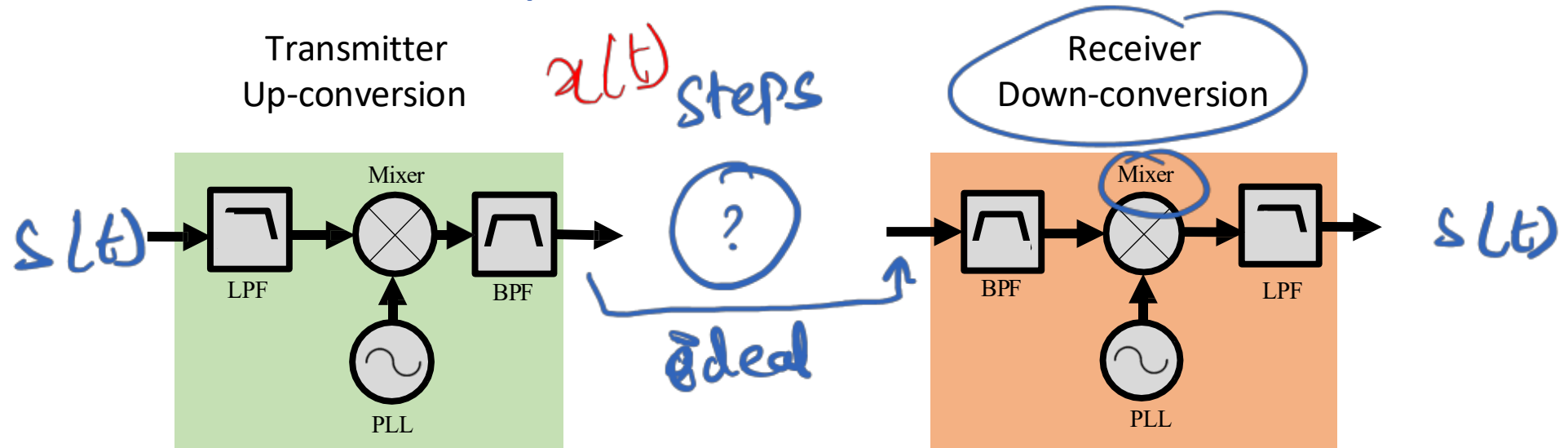
Transmitter



Receiver



Up/Down Conversion



How to we do up/down conversion?

$$s(t) \xrightarrow{\cos(2\pi f_c t)} s(t) \cos(2\pi f_c t) \xrightarrow{\cos(2\pi f_c t)} s(t) \cos^2(2\pi f_c t)$$

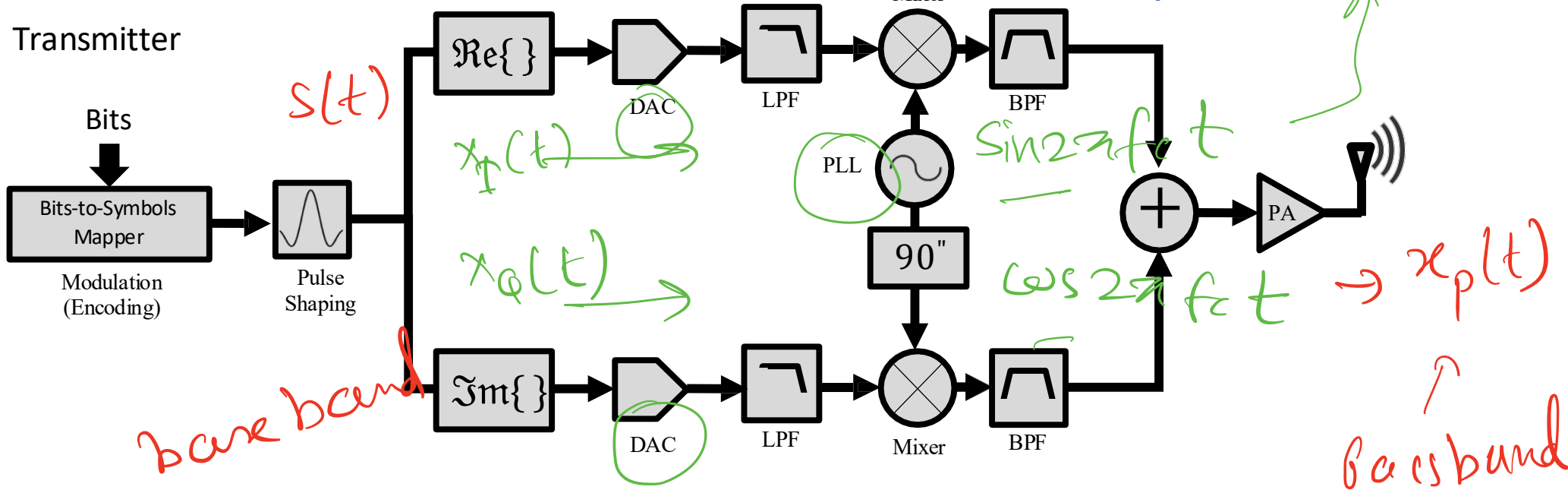
$x(t)$

$2\cos^2\theta = 1 + \cos 2\theta$

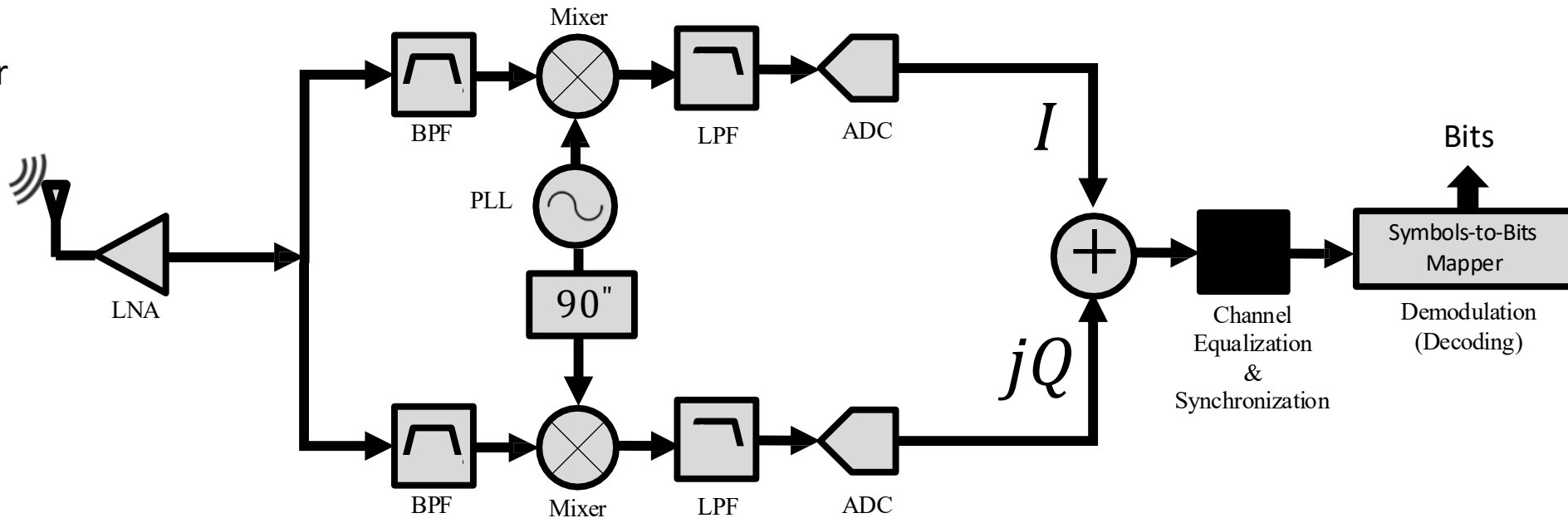
$$s(t) \cos^2(2\pi f_c t) = s(t) \left(\frac{1}{2} + \frac{1}{2} \cos(2\pi 2f_c t) \right)$$

Wireless Communication System

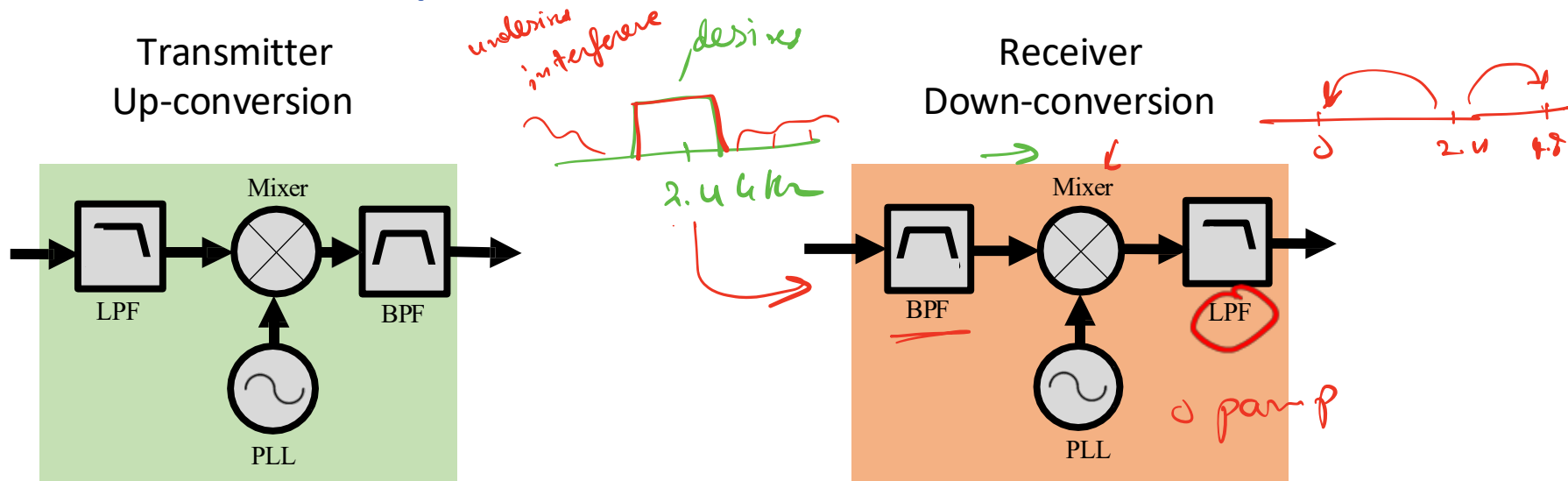
Transmitter



Receiver



Up/Down Conversion

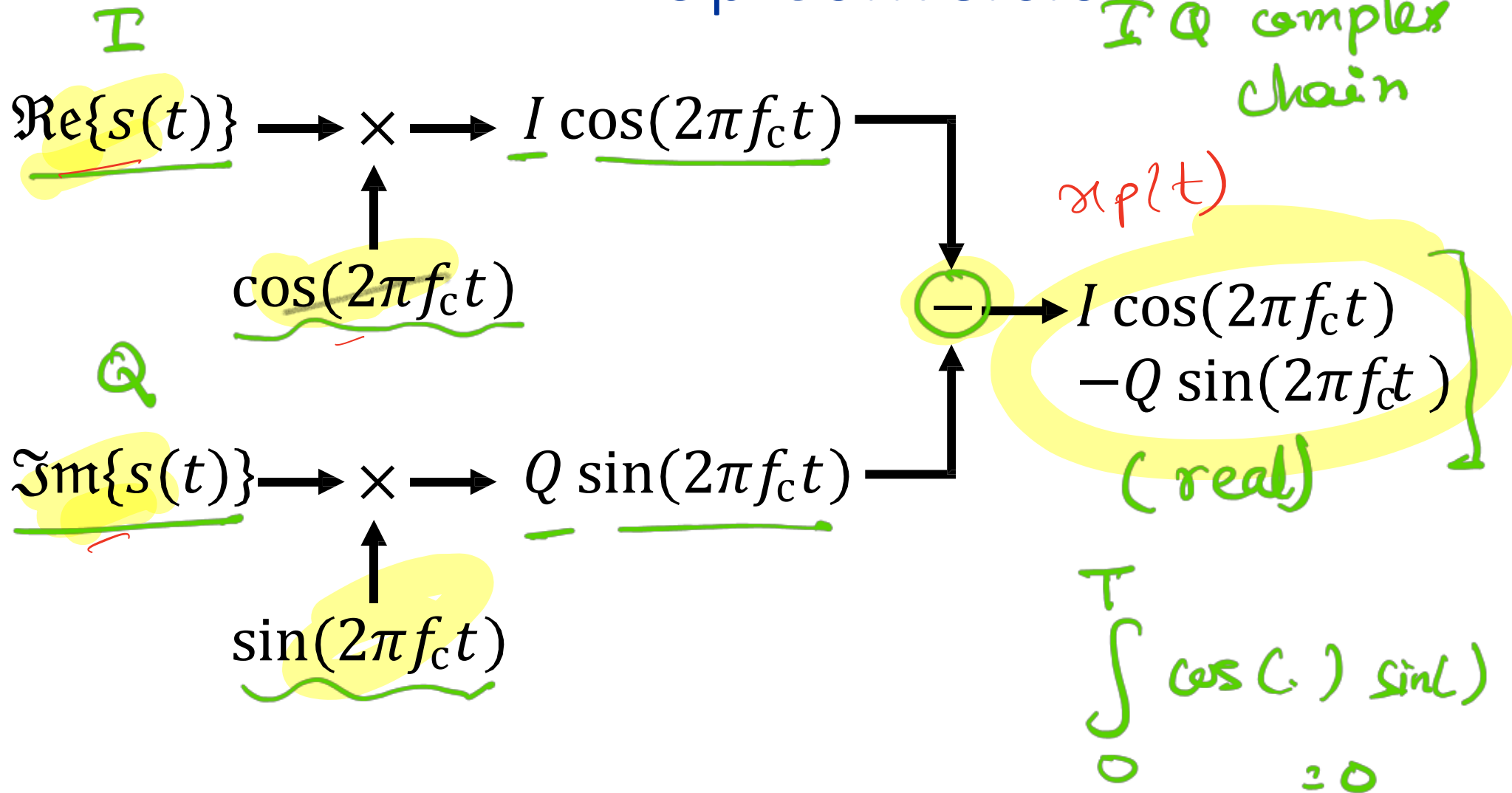


How to we do up/down conversion?

$$s(t) \xrightarrow{\uparrow \sqrt{2} \cos(2\pi f_c t)} \times \xrightarrow{\sqrt{2}} s(t) \cos(2\pi f_c t) \xrightarrow{\uparrow \frac{\sqrt{2}}{2} \cos(2\pi f_c t)} \times \xrightarrow{\sqrt{2} \times \sqrt{2} = 2} s(t) \cos^2(2\pi f_c t)$$

$$s(t) \cos^2(2\pi f_c t) = s(t) \left(\frac{1}{2} + \frac{1}{2} \cos(4\pi f_c t) \right) \xrightarrow{\text{Low Pass Filter (LPF)}} \frac{1}{2} s(t)$$

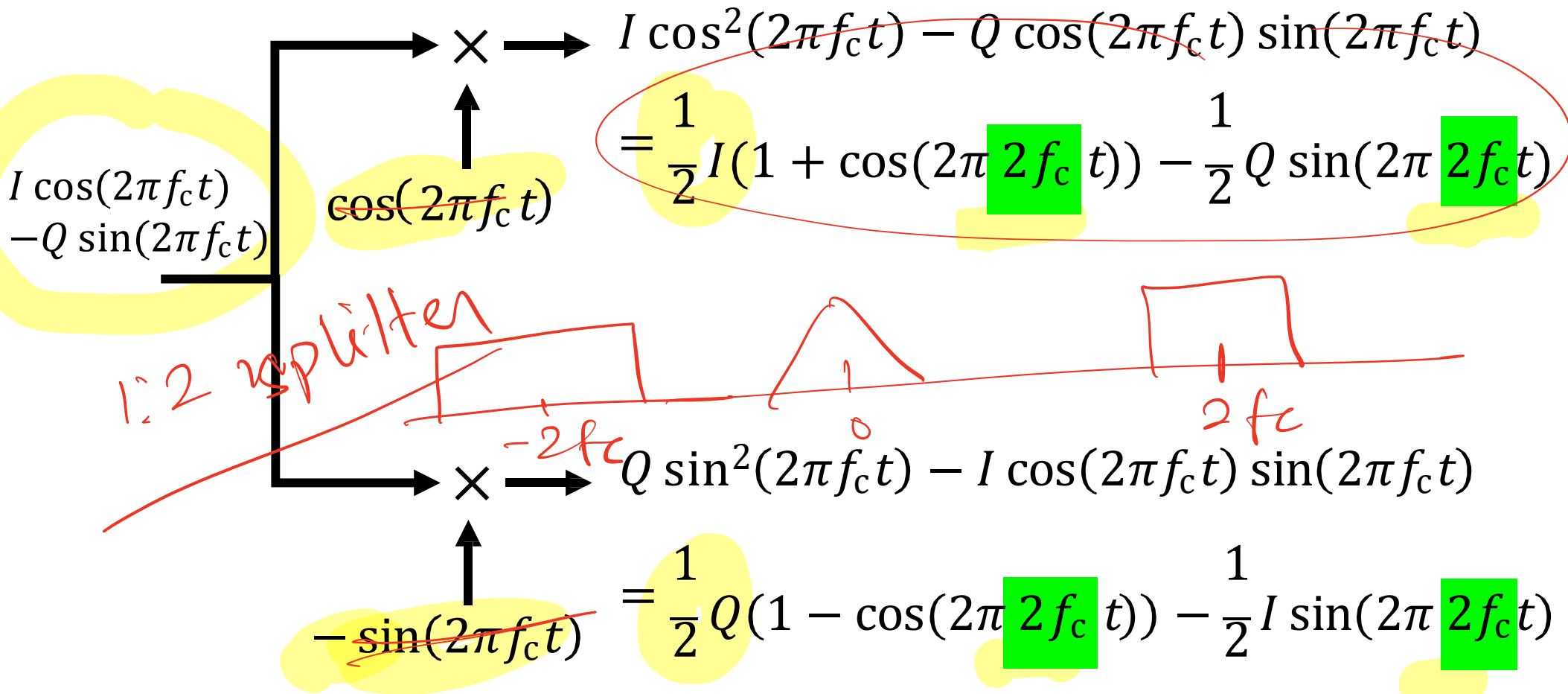
Up Conversion



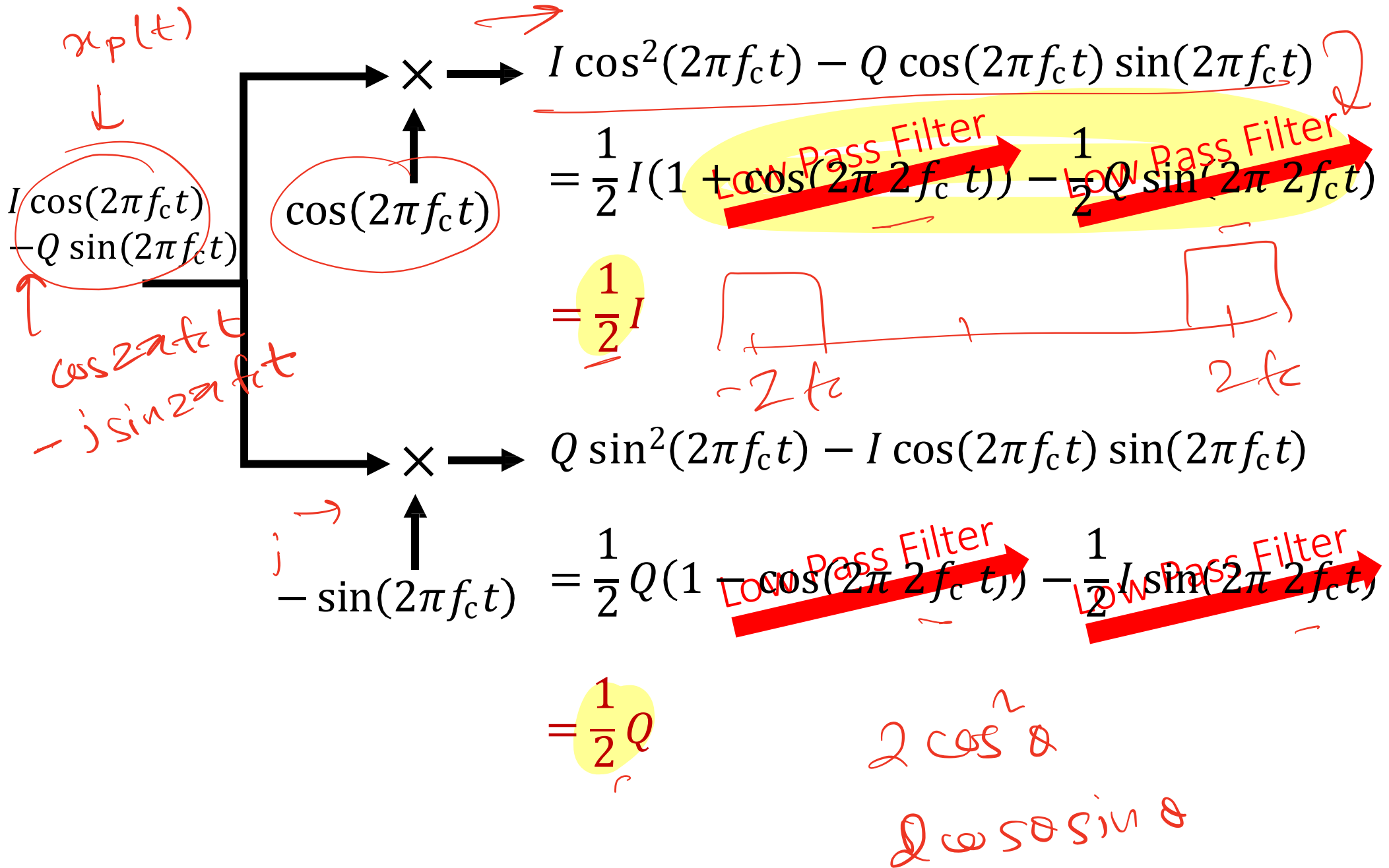
Consider using QAM: Quadrature Amplitude Modulation

$s(t)$ is complex: $s(t) = I + jQ$

Down Conversion



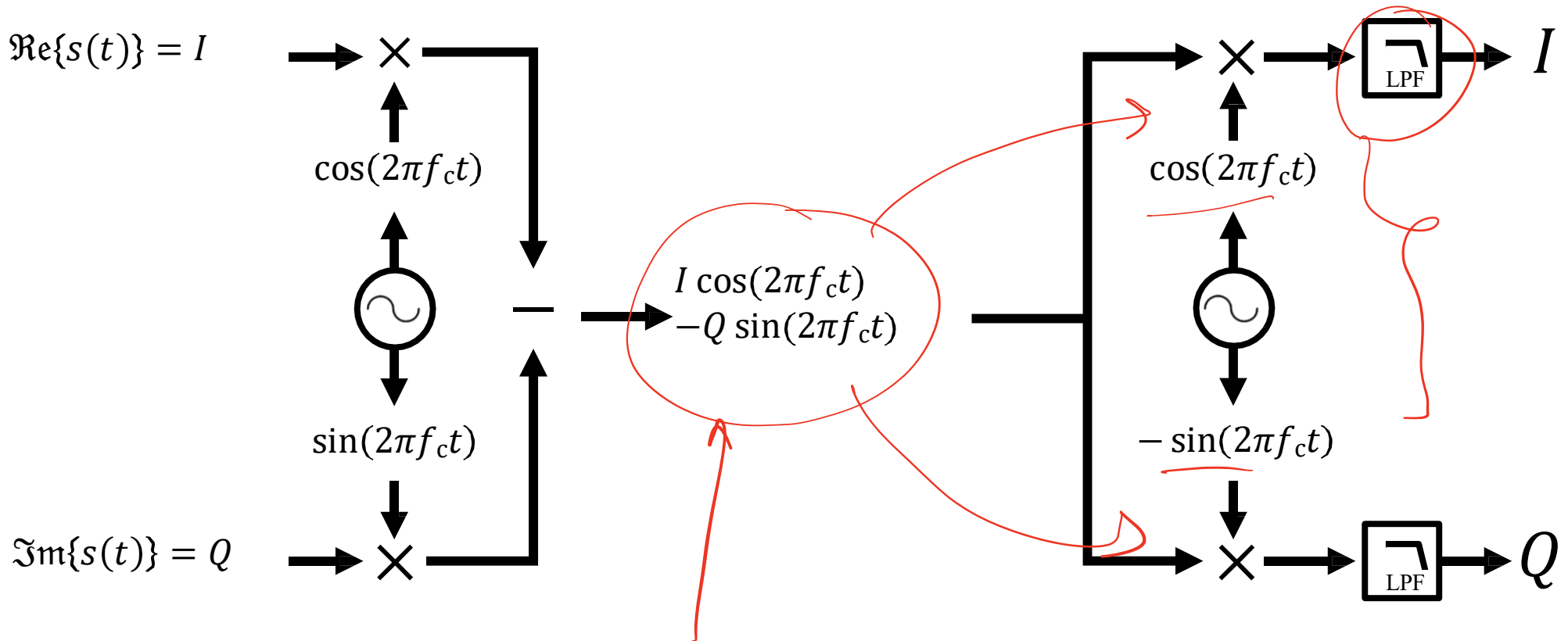
Down Conversion



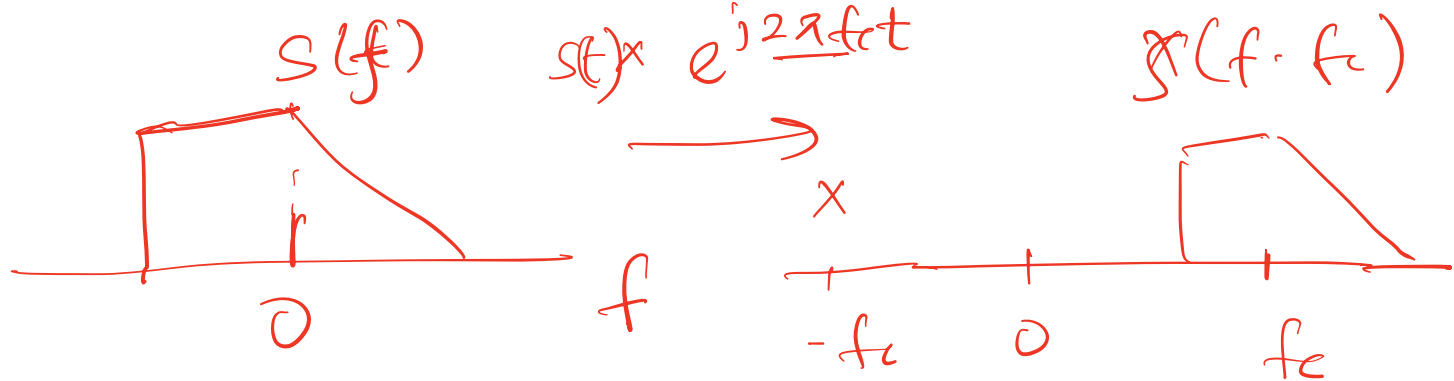
Up/Down Conversion

Transmitter

Receiver

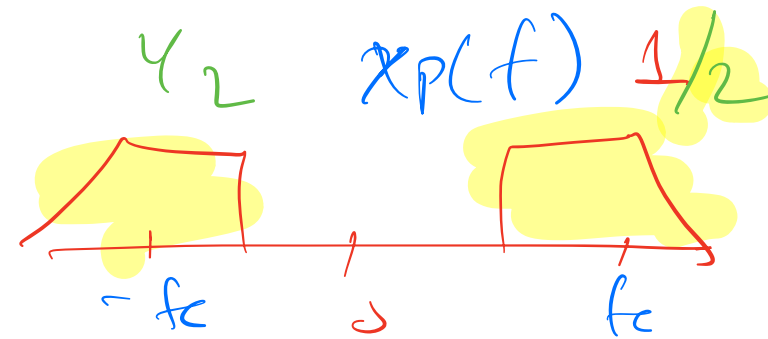


$$\cancel{s(t)} \times \cancel{e^{j2\pi f_c t}} \rightarrow \Re\{s(t)e^{j2\pi f_c t}\} \rightarrow \times e^{-j2\pi f_c t} \rightarrow \frac{s(t)}{2}$$



$$\text{Re}(a) = \frac{a + \text{conj}(a)}{2}$$

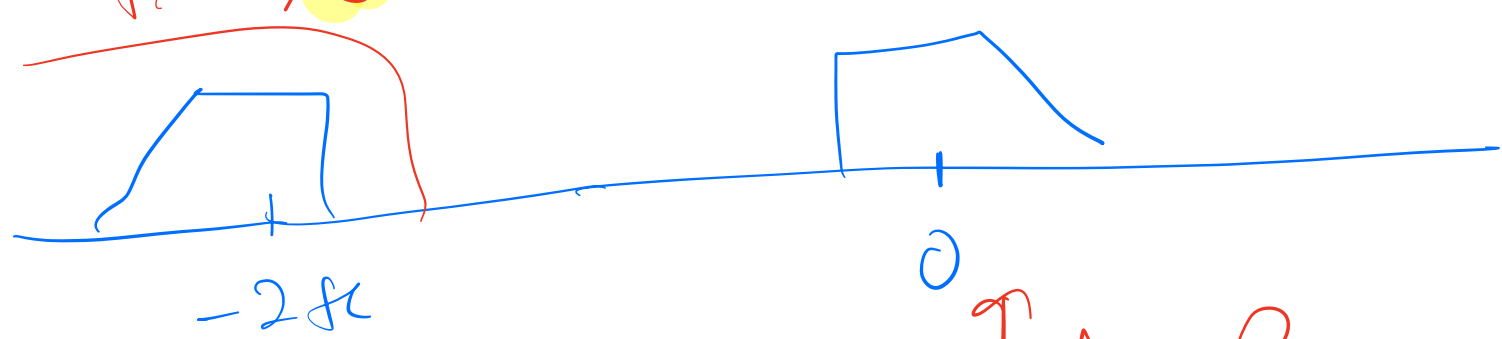
$$\downarrow \text{Re}(s(t) e^{j2\pi f_c t})$$



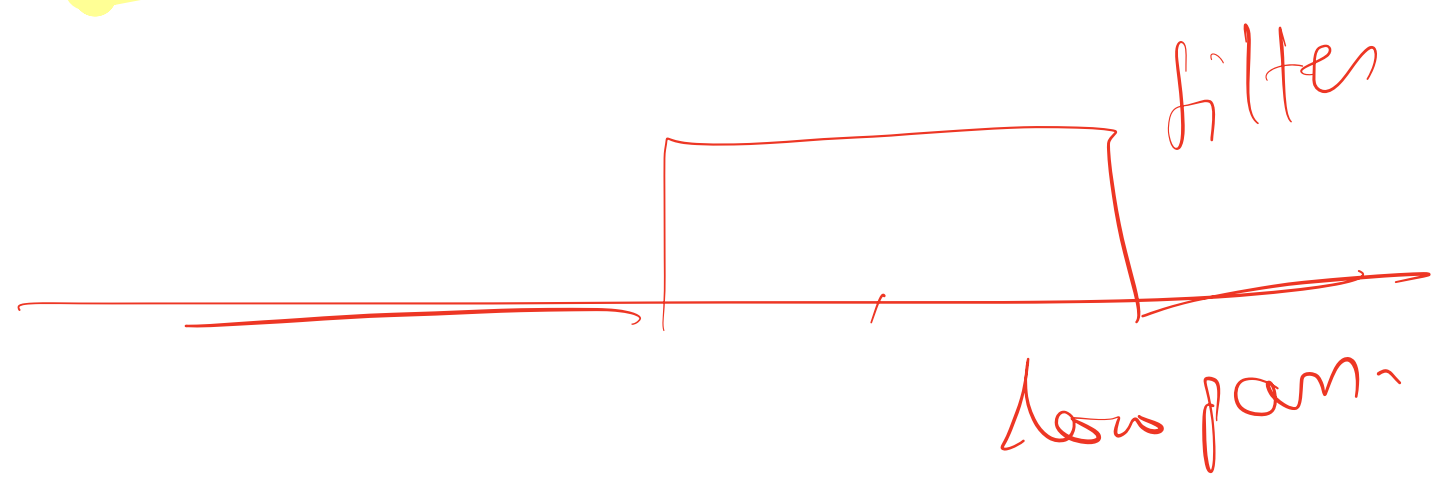
Rx operation

$$y_b(t) = x_p(t) e^{-j2\pi f_c t}$$

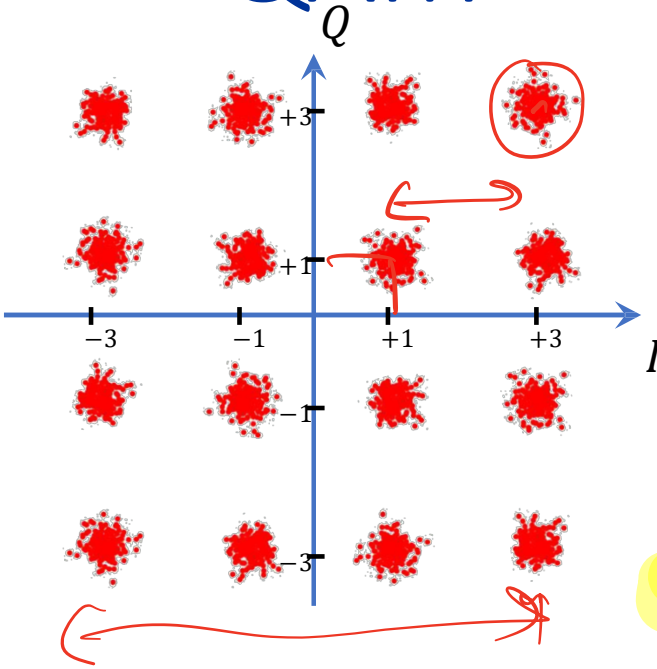
filter out



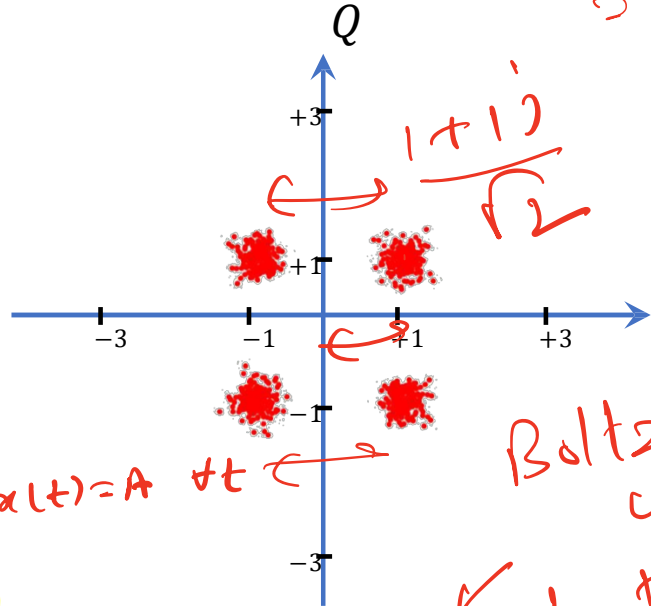
↑ keep



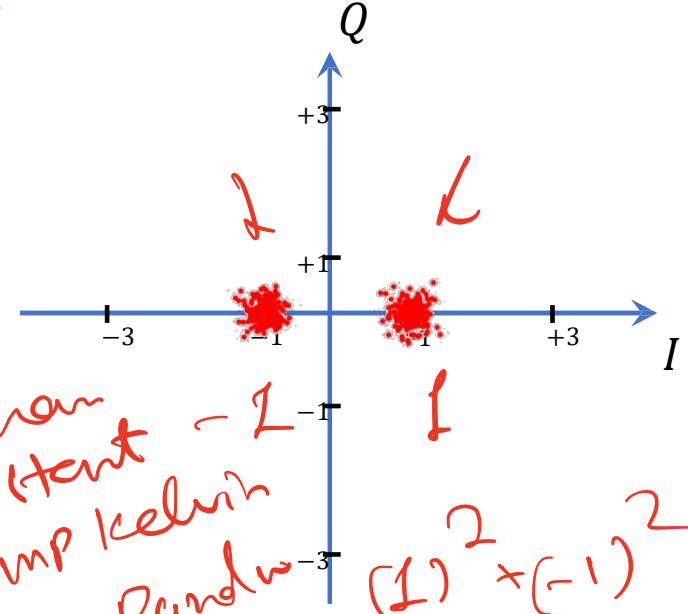
16-QAM



4-QAM



BPSK



$$SNR = \frac{E[|x(t)|^2]}{N_0}$$

$$SNR = \frac{(4 \times 2 + 8 \times 10 + 4 \times 18) / 16}{N_0}$$

$$= \frac{10}{N_0}$$

>

$$SNR = \frac{(2 + 2 + 2 + 2) / 4}{N_0}$$

$$= \frac{2}{N_0}$$

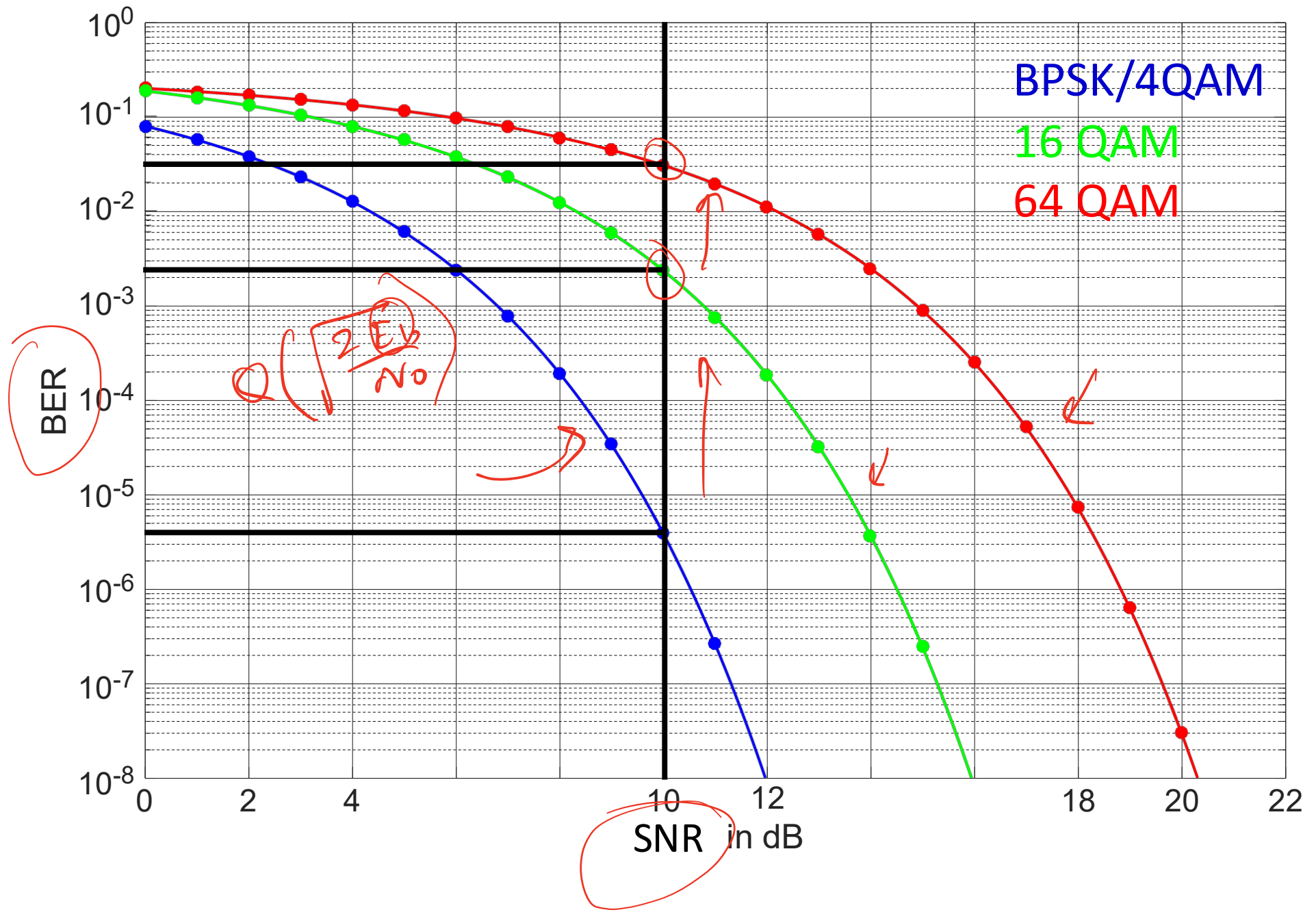
1/2

>

$$SNR = \frac{(1 + 1) / 2}{N_0}$$

$$= \frac{1}{N_0}$$

Bit Error Rate (BER)



Wireless Data Rates

• Data Rate

- Bandwidth: Samples/sec
- Modulation: Bits/sample
- Coding Rate: Data Bits/Coded Bits

$$\text{Data Rate} = \text{Bandwidth} \times \text{Bits/sample} \times \text{Code Rate}$$

sample/sec

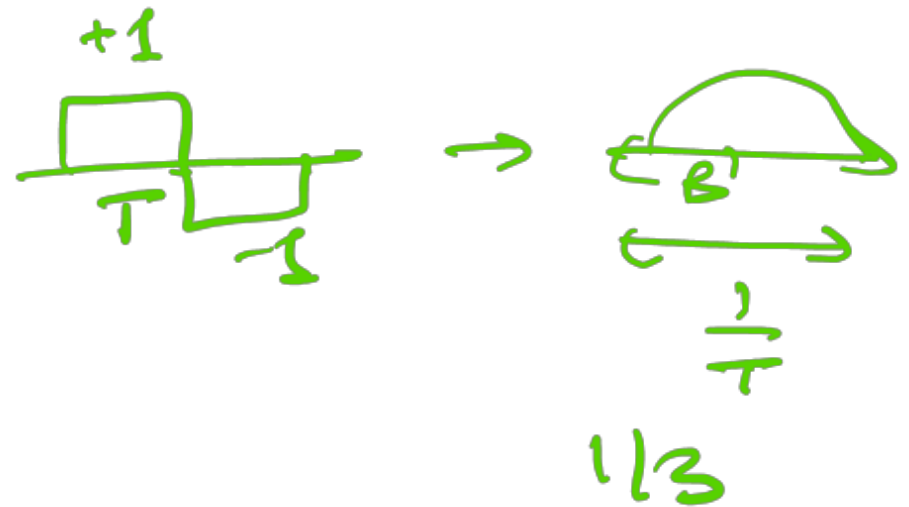
useful bits/sample

• Capacity

- Maximum Achievable Data Rate
- Shannon Capacity Theorem:

$$\text{Capacity} = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$

index of modulation



Noise for Real and Complex Modulation

Definition of BW is

real $\rightarrow$ one sided BW

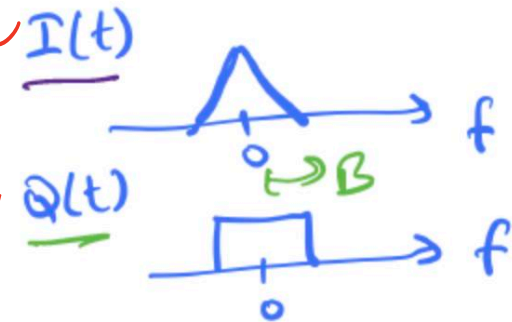
complex $\rightarrow$ two sided BW

Complex Up Conversion

$$S(t) = I + iQ$$

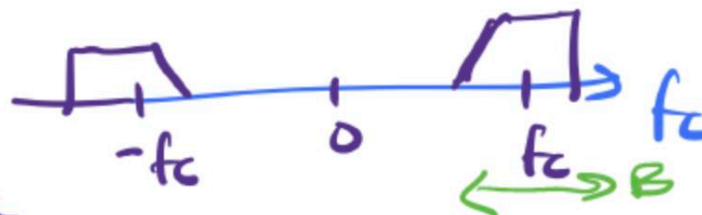
$$I \cos 2\pi f_c t$$

$$Q \sin 2\pi f_c t$$

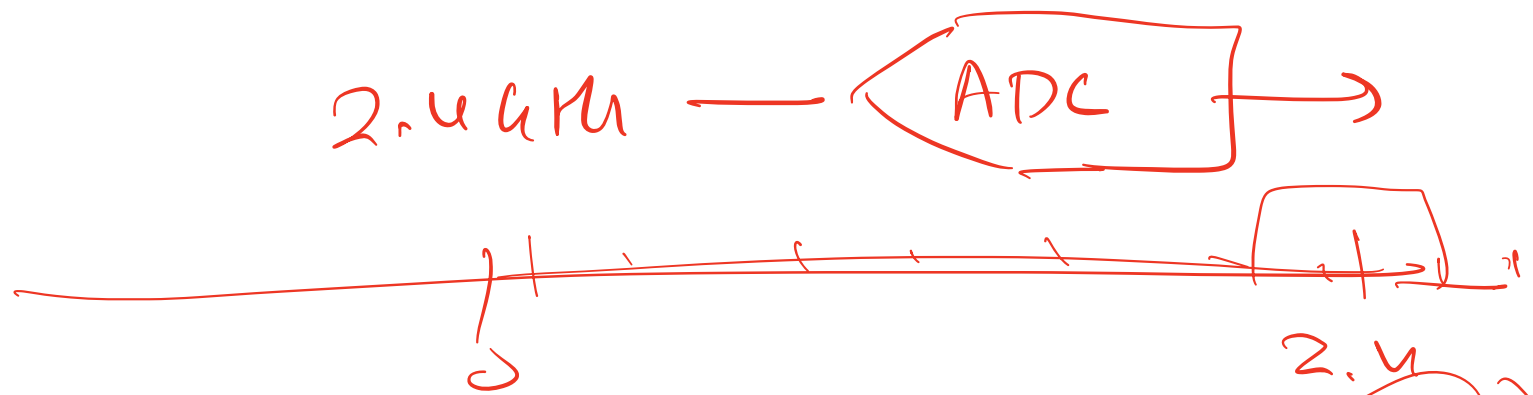


$$I(t) \cos 2\pi f_c t - Q(t) \sin 2\pi f_c t$$

real signal
produce symmetric
magnitude freq response (ideally conj. symm.)



$$f_c - \frac{B}{2} \text{ to } f_c + \frac{B}{2}$$



Question 1: Wireless Transceiver Chain and Modulation

- Explain why *up-conversion* is necessary in wireless systems. What fundamental limitation would arise if we tried to transmit at baseband frequencies?
- For a system operating at carrier frequency $f_c = 2.4$ GHz and bandwidth $B = 20$ MHz, what is the minimum sampling rate required to directly digitize the passband signal? Compare it with the rate required for complex baseband sampling.
- A transmitter uses 16-QAM with average symbol energy $E_s = 10^{-3}$ J. Sketch the constellation and compute the minimum Euclidean distance between points.
- If the same transmit power is used for BPSK, QPSK, and 16-QAM, explain qualitatively which modulation achieves the lowest BER and why.

$$\frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2\sigma^2}(\frac{y_i - \theta}{\sigma})^2}$$

Decoding under noise

Assume Data: $y_i \sim \mathcal{N}(\theta, \sigma^2)$

signal

$$= \theta + n, n \sim \mathcal{N}(0, \sigma^2)$$

MLE (maximum likelihood):

$$\hat{\theta}_{MLE} = \arg \max_{\theta} p(\bar{\mathbf{y}} | \theta)$$

Chooses the parameter that makes the observed data most likely.

MAP (maximum a posteriori):

$$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta | \mathbf{y}) = \arg \max_{\theta} \underbrace{\log p(\mathbf{y} | \theta)}_{\text{likelihood}} + \underbrace{\log p(\theta)}_{\text{prior}}$$

Bayes rule

likelihood

prior



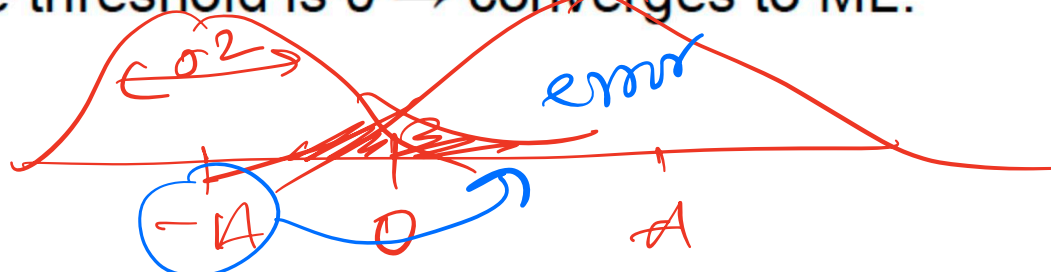
Decision Boundary

$$y = s_m + \sqrt{}$$

- Decide between BPSK symbols $s \in \{\pm A\}$ given y .
- ML detector: $\arg \max_m p(y | s_m) \rightarrow$ nearest-neighbor rule.
- **MAP detector**: $\arg \max_m p(y | s_m) P(s_m) \rightarrow$ ML plus prior on symbol probabilities.
- Binary antipodal example $y = s + \mathcal{N}(0, \sigma^2)$, $s \in \{\pm A\}$:
Decide $+A$ if

$$\underbrace{\ln \frac{p(y | +A)}{p(y | -A)}}_{(2Ay)/\sigma^2} \geq \underbrace{\ln \frac{P(-A)}{P(+A)}}_{\text{priors.}} \Rightarrow y \geq \frac{\sigma^2}{2A} \ln \frac{P(-A)}{P(+A)}.$$

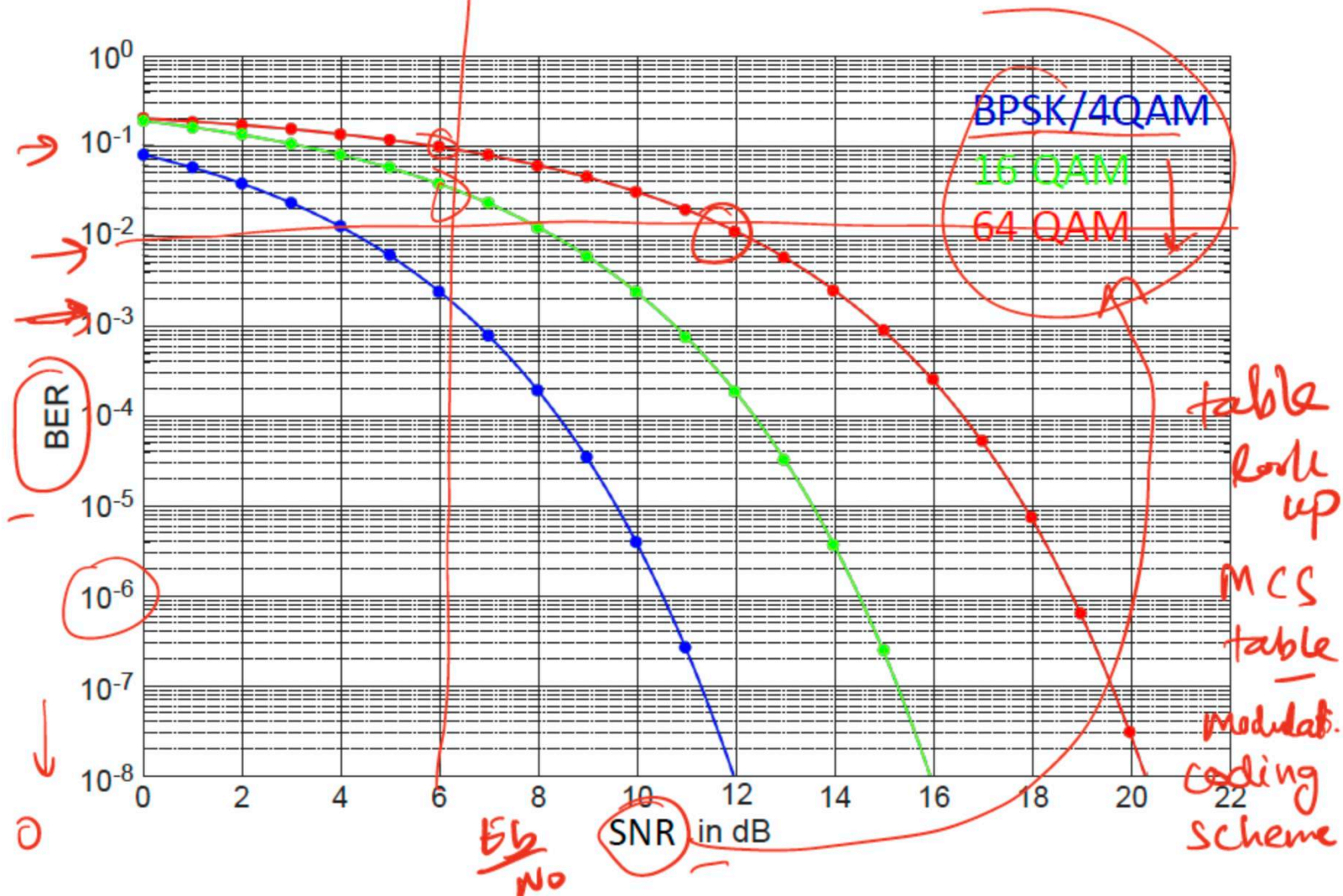
- With equal priors the threshold is 0 $\rightarrow$ converges to ML.



Derive BER

Monte Carlo
Simulation.

Bit Error Rate (BER)



BPSK

Bit Error Rate (BER)

Energy per bit
 $= \frac{(A)^2 + (-A)^2}{2}$
 $= E_b$



$P(\text{bit error})$
 $= P(y_i > 0 | H_0) \text{ or } P(y_i < 0 | H_1)$
 $= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(y_i - (-A))^2}{\sigma^2}} = P(y_i | -A)$

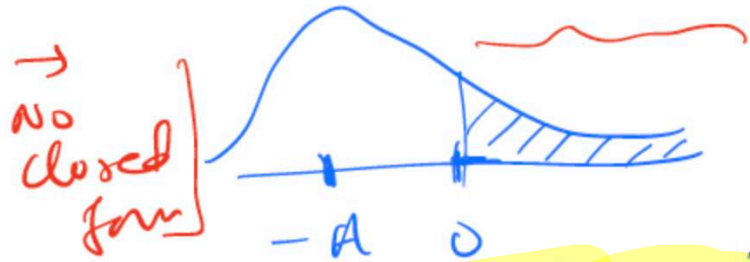
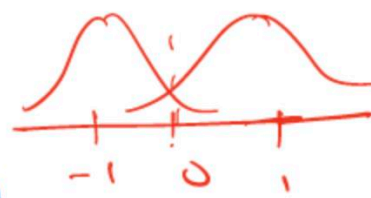
$P(-A + v > 0)$
 $= P(v > A) = P\left(\frac{v}{\sigma} > \frac{A}{\sigma}\right)$

$\frac{v}{\sigma} \sim N(0, 1)$

Q function: $P(\text{st-normal dist} > a)$
 $Q(a)$

$P(v > A)$
 $= Q\left(\frac{A}{\frac{\sigma}{\sqrt{2}}}\right)$
 $= Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$ where $\frac{E_b}{N_0} = \text{SNR}$

$P_e(1|-1)$
 $P_e(-1|1)$



$Q(a) = P(y > a)$
 $y \sim N(0, 1)$

$P(v > A)$
 $= P\left(\frac{v}{\sigma} > \frac{A}{\sigma}\right)$
 $= Q\left(\frac{A}{\frac{\sigma}{\sqrt{2}}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

$A = \sqrt{E_b}$
 $\sigma^2 = \frac{N_0}{2}$

$Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

Question 2: Estimation and Decision Theory

0 $\rightarrow +A$
1 $\rightarrow -A$

- (a) Consider a binary detection problem $y = s + n$, where $s \in \{+A, -A\}$ and $n \sim \mathcal{N}(0, \sigma^2)$. Derive the Maximum Likelihood (ML) decision rule and the corresponding decision boundary.
- (b) Modify the result for the MAP detector if $P(s = +A) = 0.8$ and $P(s = -A) = 0.2$. What happens to the threshold?
- (c) Using the Q -function, derive the expression for bit-error probability

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

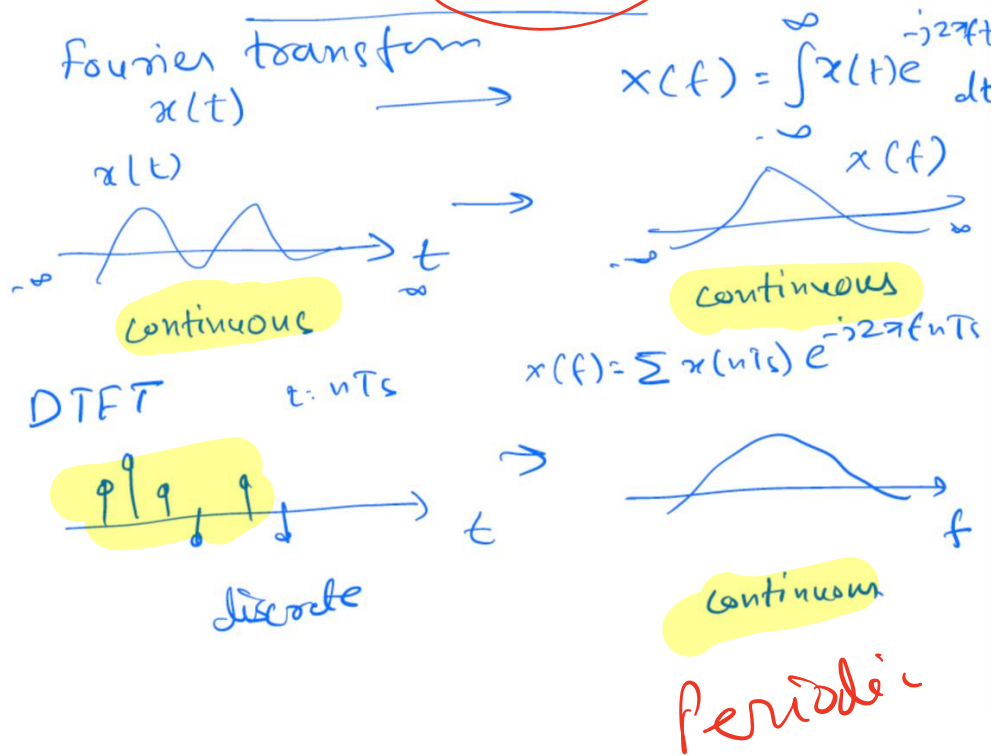
for BPSK over AWGN.

0 0 0 0 0 0 0
0 1

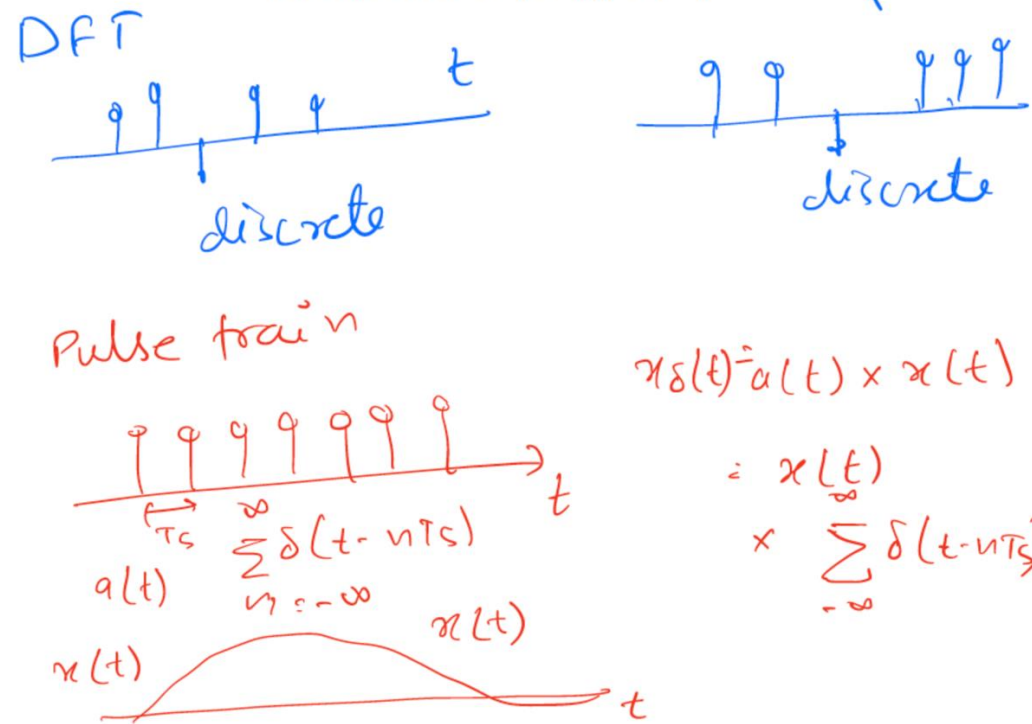
- (d) How would the performance change if we use 16-QAM instead of BPSK at the same E_b/N_0 ? Explain using constellation geometry and SNR per bit reasoning.

DSP

Review: DTFT, DFT



Review: DTFT, DFT



Ideal Sampling

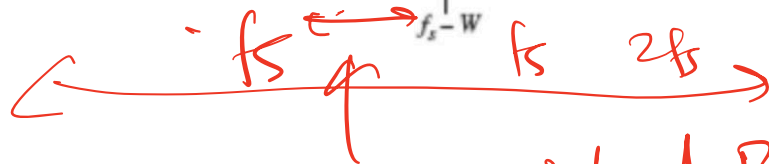
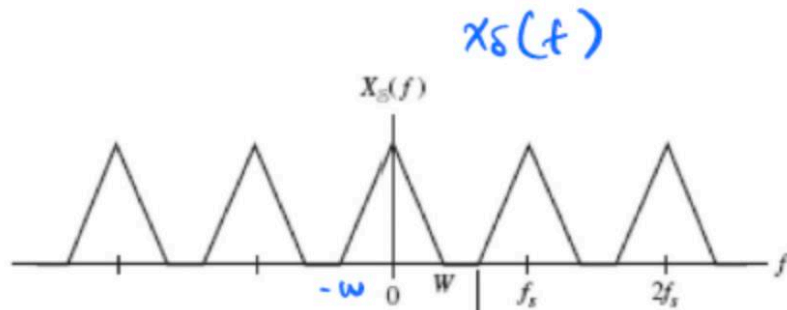
Ideal sampling

Sampling using "delta" functions

$$\begin{aligned} x_\delta(t) &= \sum_{k=-\infty}^{\infty} x(kT_s) \delta(t - kT_s) \\ &= x(t) \sum_{k=-\infty}^{\infty} \delta(t - kT_s) \end{aligned}$$

Spectrum

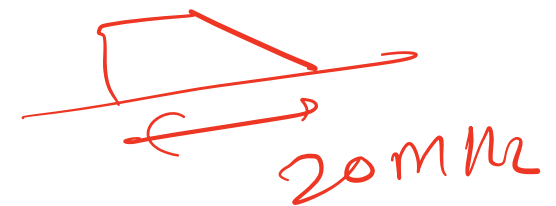
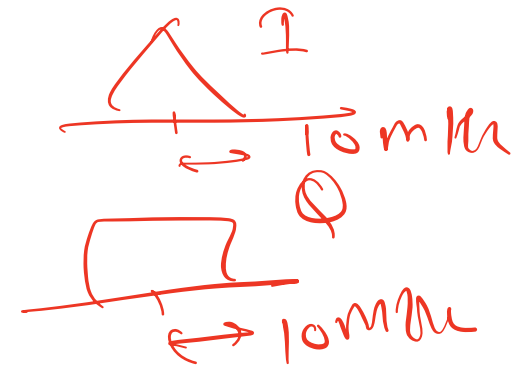
$$X_\delta(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s) \quad f_s = \frac{1}{T_s}$$



onesided BW = $B = W$

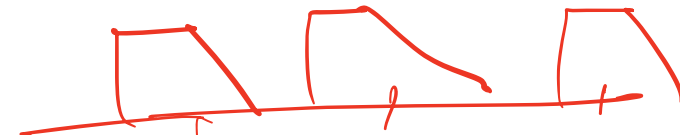
twosided BW = $2W$

ADC



Nyquist

$$f_s \geq 2W$$



Ideal reconstruction (3/3)

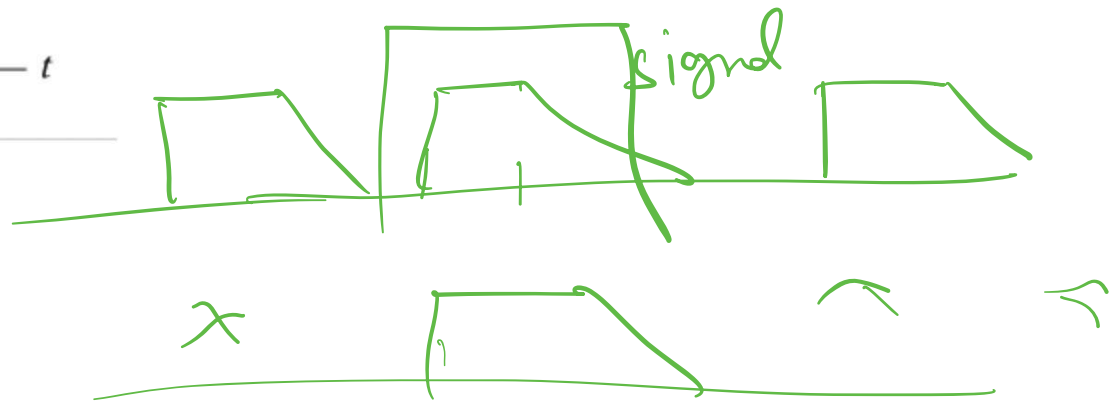
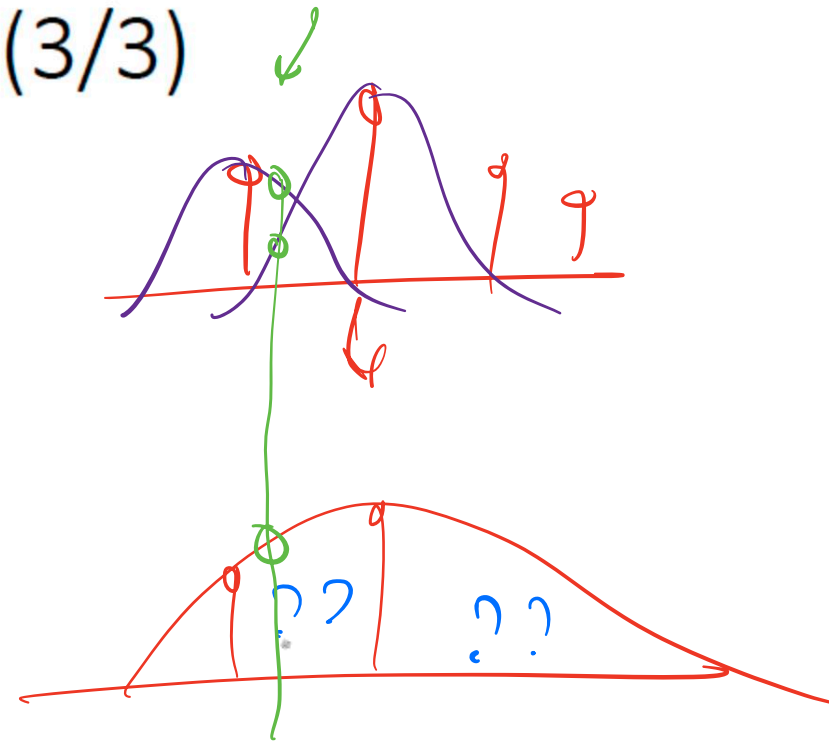
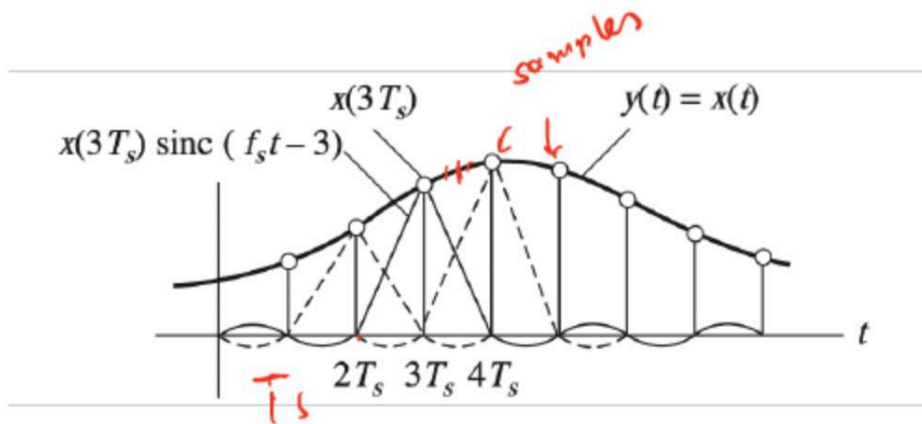
Reconstruction by interpolation (time domain)

Reconstruct from samples, but with ideal interpolation

$$h(t) = \mathcal{F}^{-1}[H(f)] = 2BK \text{sinc} 2B(t - t_d)$$

$$\Rightarrow y(t) = h(t) * x(kT_s) = 2BK \sum_{k=-\infty}^{\infty} x(kT_s) \text{sinc} 2B(t - t_d - kT_s)$$

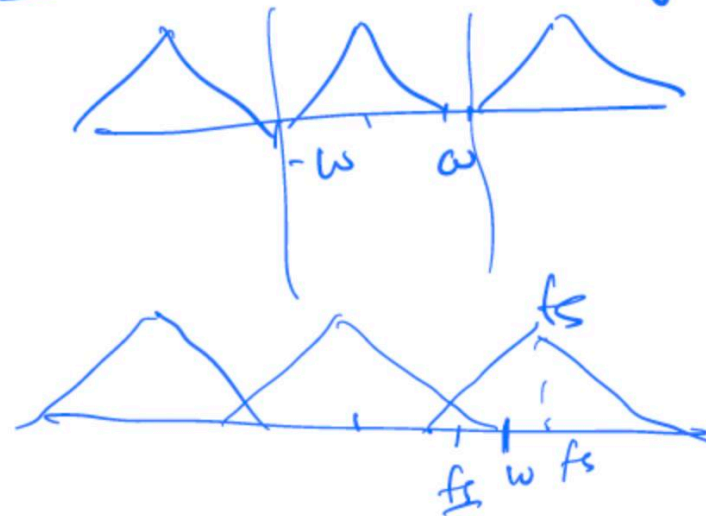
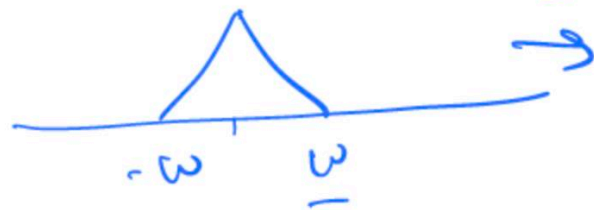
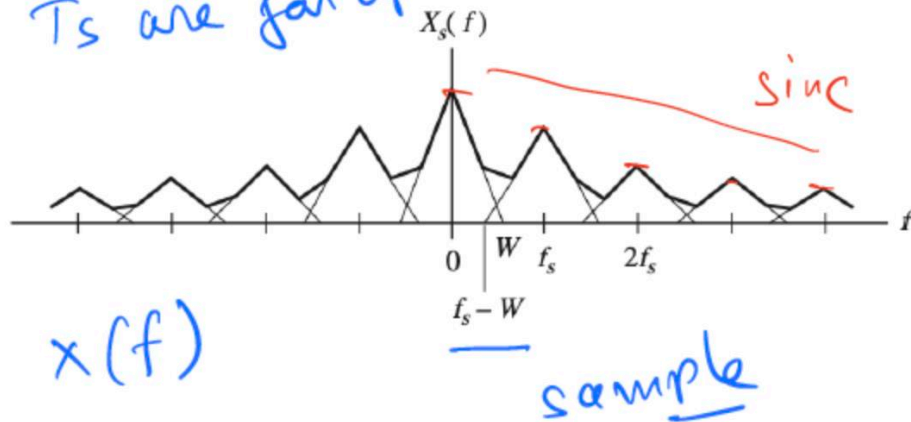
(Unimplementable in practice)



Aliasing

Under-sampled message

T_s are far apart



$$f_s \geq 2w$$

No aliasing

$$f_s < 2w$$

aliasing

Question 3: Sampling and Reconstruction

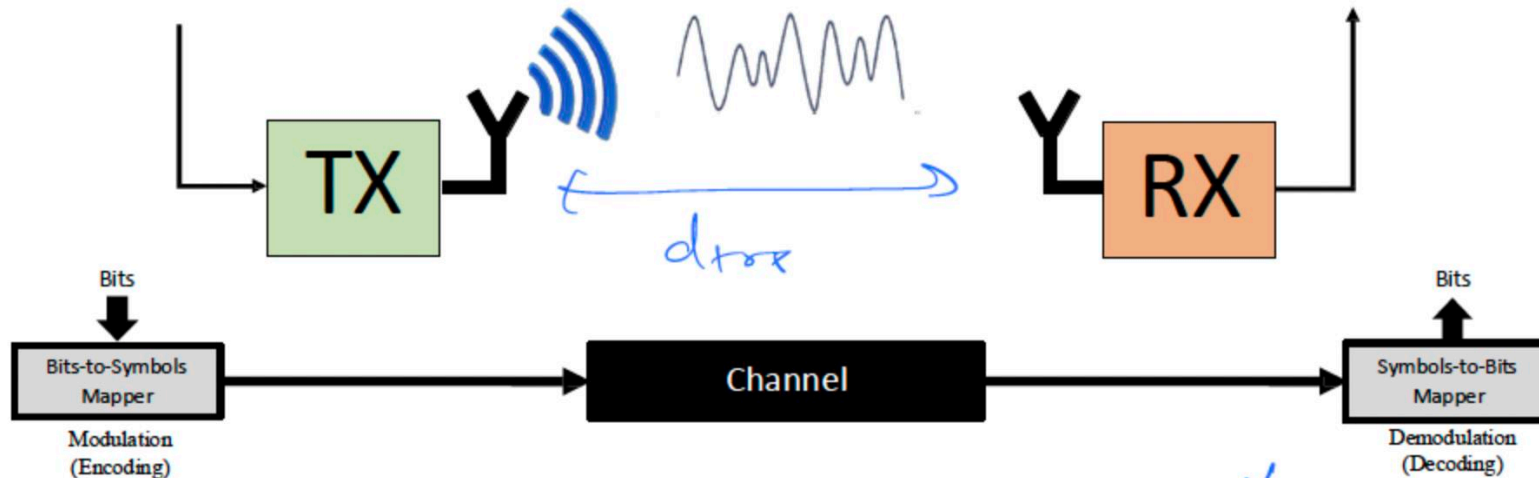
- (a) State the Nyquist Sampling Theorem. Explain what happens when the sampling rate is below, equal to, and above $2B$, where B is the signal bandwidth.
- (b) A continuous-time baseband signal $x(t)$ with maximum frequency 3 MHz is sampled at 4 MHz. Sketch its sampled spectrum and explain the aliasing that occurs.
- (c) Describe ideal reconstruction using an LPF. Provide the expression for the reconstructed signal in terms of sinc interpolation.

The Channel



1011010110011001

1011010110011001



Handwritten notes: $\lambda = \frac{c}{f}$ and $f = \frac{c}{\lambda}$. The signal $x(t)$ is shown entering the channel.

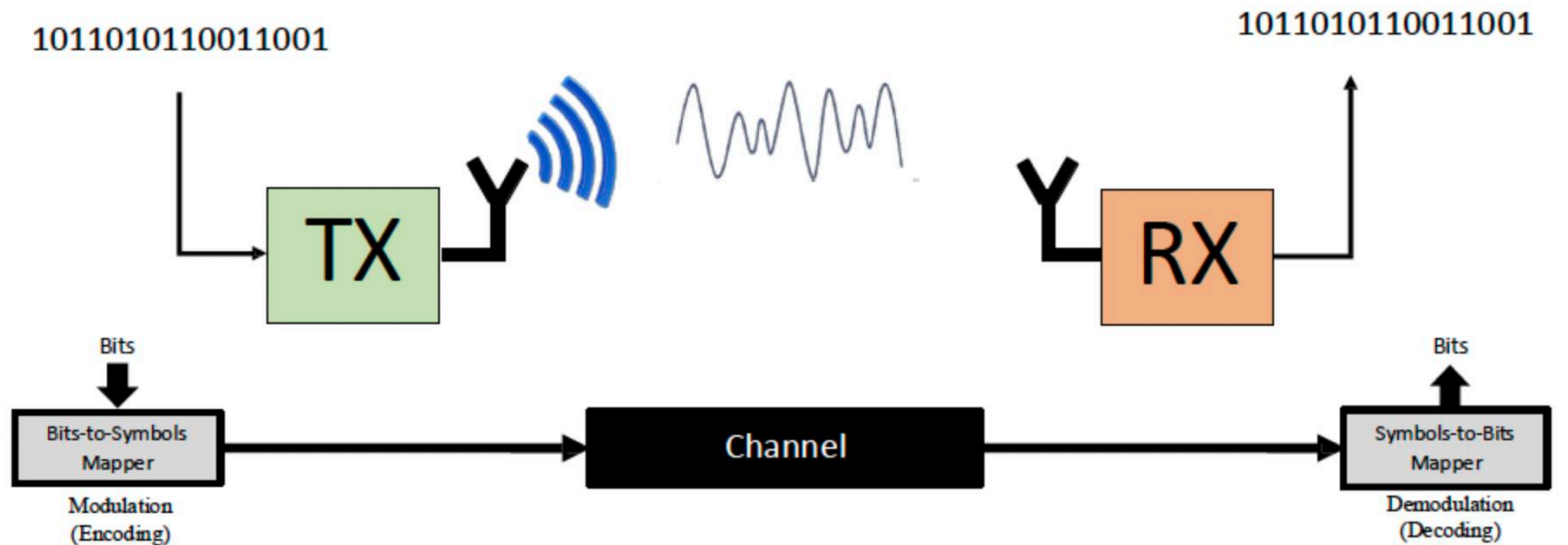
The received signal is given by: $y(t) = \mathbf{h} x(t - \tau) + v(t)$

Channel attenuates the signal (Pathloss)

Handwritten notes: P_{RX} is underlined, G_{TX} and G_{RX} are labeled 'TX antenna gain' and 'RX antenna gain', and P_{TX} is labeled 'TX power'. The distance d_{TX} is also indicated.

$$P_{RX} = G_{TX} G_{RX} \frac{\lambda^2}{(4\pi d)^2} P_{TX} \Rightarrow |h| \propto \frac{\lambda}{d}$$

The Channel



$$x(t) \Rightarrow$$

$$\Rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

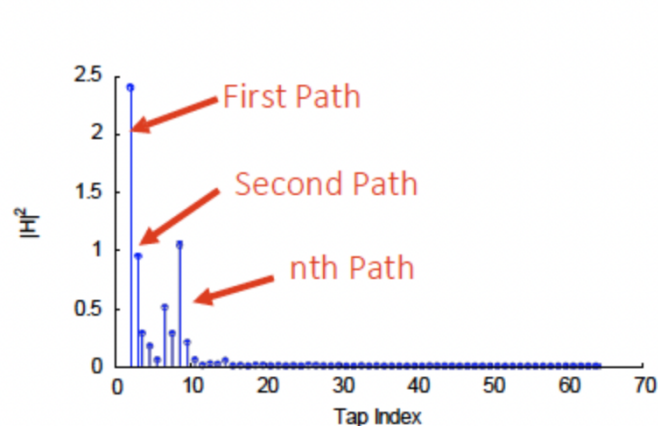
$$x(t) \times e^{-j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{-j2\pi f_c (t - \tau)} \Rightarrow \times e^{j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{j2\pi f_c t}$$

$$h \propto \frac{\lambda}{d} e^{j\phi} \rightarrow \phi = 2\pi f_c \tau = 2\pi \frac{c}{\lambda} \frac{d}{c} = 2\pi \frac{d}{\lambda} \rightarrow h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

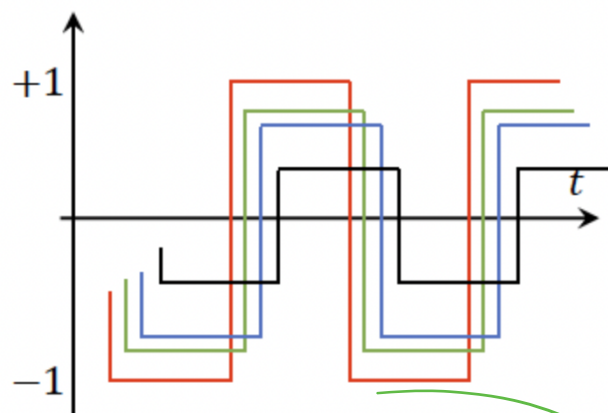
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



ISI: Inter-Symbol-Interference
Symbols arriving along late paths interfere with following symbols.

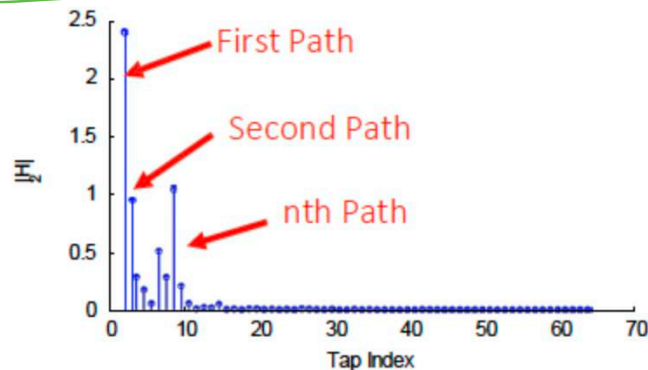
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

$\Leftrightarrow$

$$H(f)X(f)$$

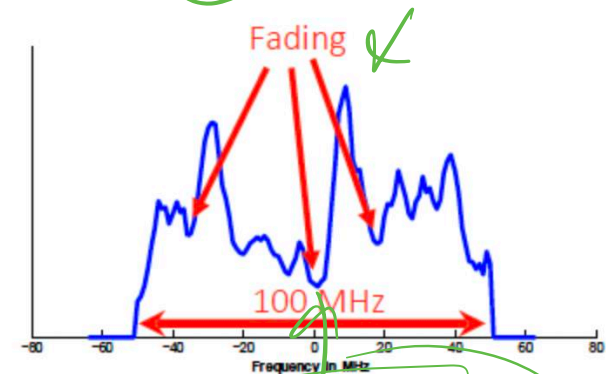


Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Frequency Selective Fading

Symbols arriving along different paths sum up destructively

Problematic in Wideband Channel!

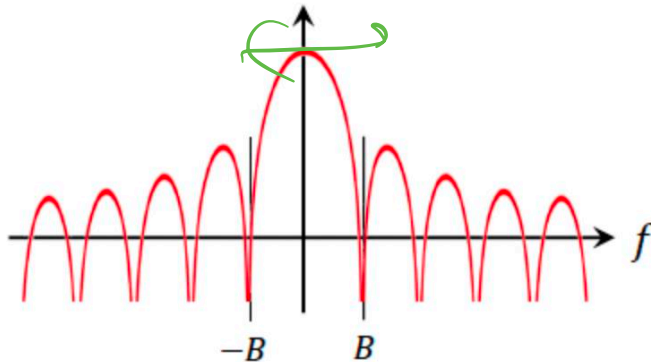
Question 4: Wireless Channel Effects

- (a) Express the general received signal model $y(t) = h(t) * x(t) + v(t)$. List and describe the four key effects introduced by the channel: noise, delay, attenuation, and phase rotation.
- (b) Using the free-space path loss formula, compute the received power in dBm at 10 m for $f = 5$ GHz and $P_T = 0$ dBm. Assume Omni-directional antennas.
- (c) Explain and illustrate multipath fading with two propagation paths of lengths 1 m and 1.06 m at 5 GHz. Calculate the phase difference and discuss constructive vs. destructive interference.
- (d) Define channel impulse response (CIR) and show how it leads to inter-symbol interference (ISI).
- (e) Describe a simple preamble-based channel estimation and equalization scheme for BPSK and explain how it restores coherence at the receiver.

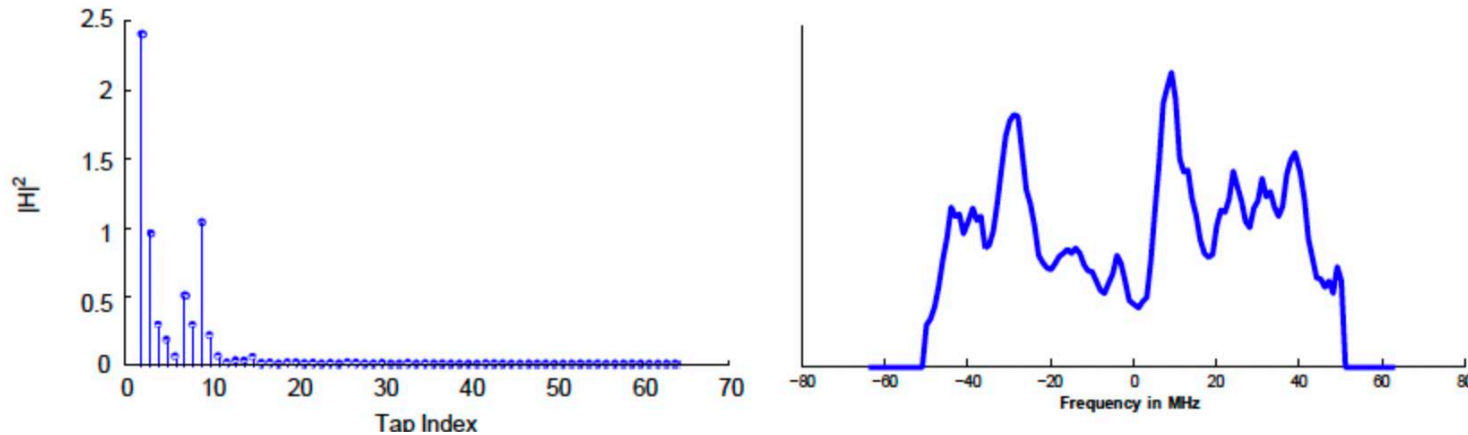
Single Carrier Modulation

Symbols modulated on a single carrier frequency

- Low Spectral Efficiency: sinc & raised cosine leakage



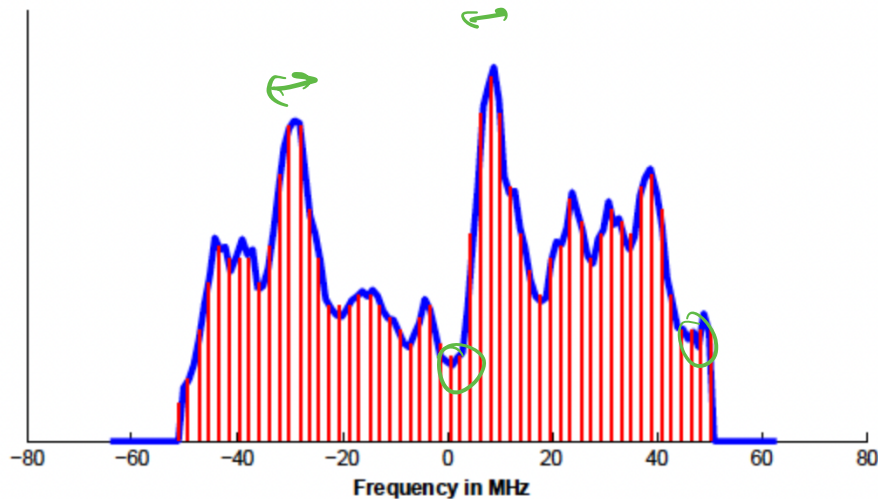
- ISI: Inter-Symbol-Interference limits performance



Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



$$x(t) = \sum_i s_i[n] e^{-j 2 \pi f_i t}$$

$$y(t) = \sum_i h_i s_i[n] e^{-j 2 \pi f_i t}$$

- Transmit symbols on different carriers in narrow bands
- Channel is Flat → No need to worry about ISI

Not That Simple!

OFDM: Orthogonal Frequency Division Multiplexing

Use DFT: Discrete Fourier Transform

$$\text{N-Point DFT: } X(f_i) = \frac{1}{N} \sum_{t=0}^{N-1} x(t) e^{-j \frac{2\pi f_i t}{N}}$$

Discrete → Discrete

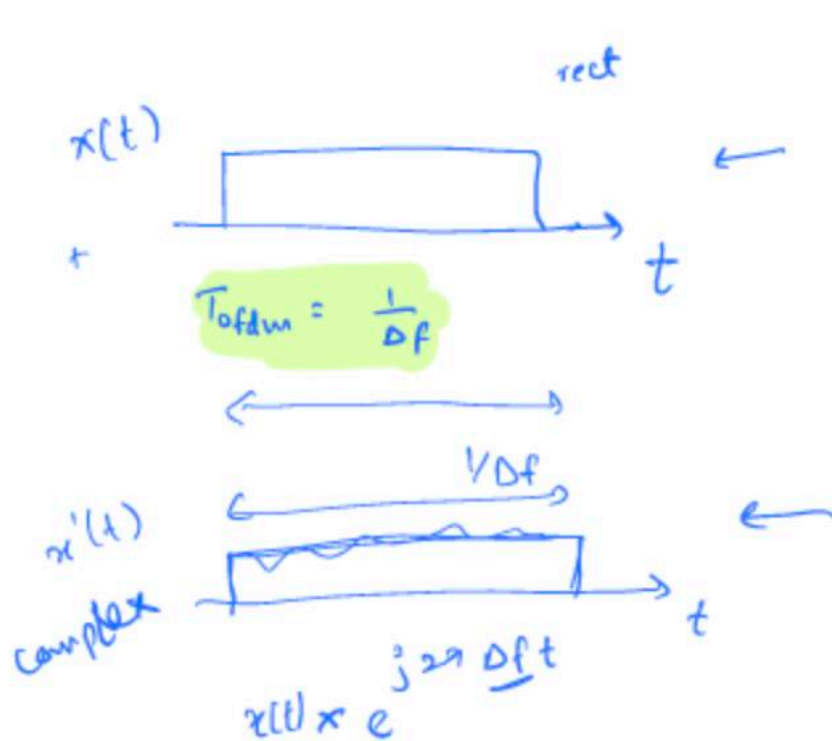
$$\text{N-Point IDFT: } x(t) = \sum_{f_i=0}^{N-1} X(f_i) e^{j \frac{2\pi f_i t}{N}} = x(f_0) e^{j \frac{2\pi f_0 t}{N}} + x(f_1) e^{j \frac{2\pi f_1 t}{N}} + \dots$$

Handwritten notes: $x(t) \rightarrow \text{DFT} \rightarrow X(f)$, $x(t) \leftarrow x(f_0)$, $x(f) = x(f_0) e^{j \frac{2\pi f_0 t}{N}} = x'(t)$

Send symbols in Frequency Domain

$$X(f_i) = s[n] \rightarrow \text{Compute and transmit } x(t) \text{ using IDFT}$$

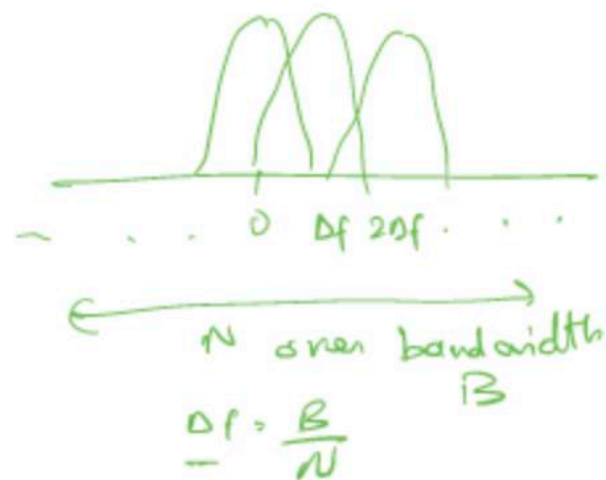
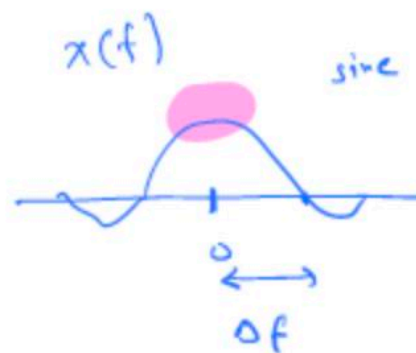
Handwritten notes: $s[n] \in \{A, -A\}$, BPSK



$$x(t) \neq x'(t) \dots \dots$$

$$T_{\text{ofdm}} = \frac{1}{\Delta f}$$

$$= \frac{N}{B}$$



OFDM Cyclic Prefix



- DFT (FFT) assumes time samples are periodic of period N
- Circular Shift before taking FFT:

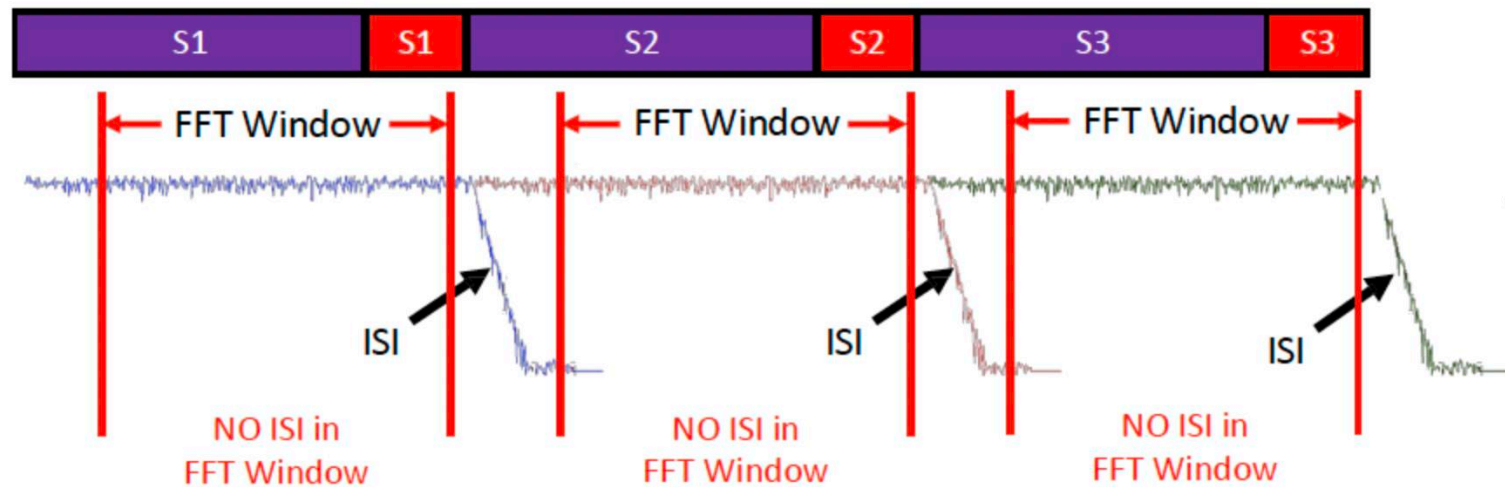
$$x[t] \rightarrow X[f]$$

$$\underline{x[t - \tau \bmod N]} \rightarrow \underline{X[f]e^{-j\frac{2\pi f\tau}{N}}}$$

OFDM Cyclic Prefix

Cyclic Prefix:

- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference



OFDM Cyclic Prefix

Cyclic Prefix:

+ Preserves orthogonality by allowing some misalignment in FFT Window

+ Deals with Inter-Symbol-Interference

— Overhead: Send $CP + N$ samples for every N samples

$$\text{Overhead} = \frac{CP}{CP + N}$$

e. g. WiFi 802.11n: $N = 64$, $CP = 16 \rightarrow \text{Overhead} = 20\%$

e. g. LTE: $N = 1024$, $CP = 72 \rightarrow \text{Overhead} = 6.5\%$

Question 5: Impact of Multipath on OFDM Orthogonality

An OFDM system uses $N = 64$ subcarriers and a sampling rate of $f_s = 20$ MHz. The cyclic prefix (CP) length is $N_{\text{CP}} = 16$ samples. The channel has two multipath components given by:

$$h(t) = \delta(t) + 0.8 e^{-j\pi/3} \delta(t - \tau_d),$$

where the delay spread is $\tau_d = 0.9 \mu\text{s}$.

- (a) Compute the OFDM symbol duration *with* and *without* CP in μs , and determine the corresponding subcarrier spacing in kHz. Briefly explain how delay spread relates to frequency selectivity in wireless channels.
- (b) Determine whether the given delay spread τ_d fits within the CP duration. If not, estimate how many time-domain samples of the delayed path extend beyond the CP. Discuss qualitatively how this affects subcarrier orthogonality and inter-symbol interference (ISI).
- (c) Write the frequency-domain response of the two-tap channel for subcarrier k :

$$H(f_k) = 1 + 0.8 e^{-j(2\pi f_k \tau_d + \pi/3)}.$$

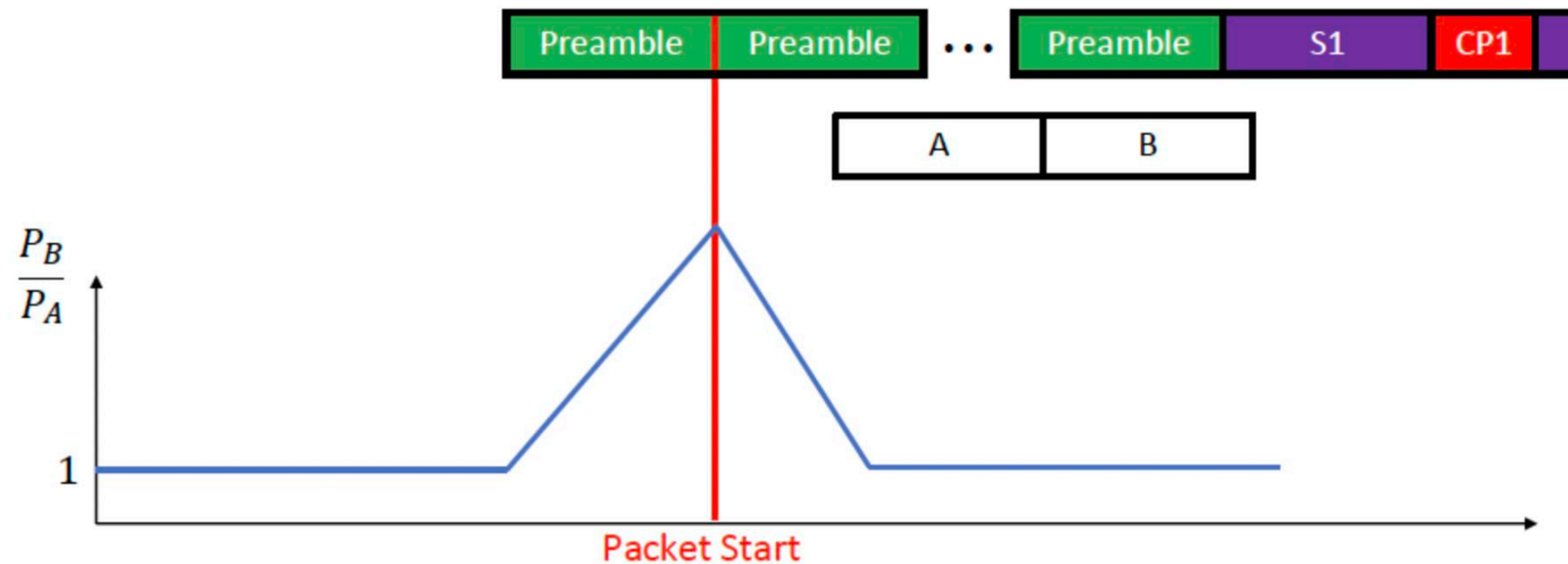
Sketch the magnitude of $|H(f_k)|$ across subcarriers, and indicate regions (values of f_k) where constructive or destructive interference occurs (frequency-selective fading).

Question 6: OFDM Design Tradeoffs

Consider an OFDM system with occupied bandwidth $B = 5$ MHz, number of subcarriers $N = 512$. Maximum channel delay spread $T_d = 4 \mu\text{s}$. Carrier frequency $f_c = 2.4$ GHz. Assume no guard bands or DC null; sampling rate $f_s = B$ and all N subcarriers carry data.

- (a) For a *single-carrier* system occupying the same bandwidth B , what is the symbol time T_{SC} ?
- (b) For the OFDM system, find the sample time T_s and the *raw* (useful) symbol time T_u (i.e., without CP).
- (c) What is the *minimum* number of CP samples $N_{\text{CP,min}}$ required? Also report the corresponding minimum CP time $T_{\text{CP,min}}$.
- (d) If the CP length is chosen as twice the minimum, i.e., $T_{\text{CP}} = 2T_{\text{CP,min}}$, what is the *total* OFDM symbol time $T_{\text{sym}} = T_u + T_{\text{CP}}$?
- (e) What is the efficiency loss in percentage due to CP overhead?
- (f) If 16-QAM is employed and all subcarriers carry data, what is the *effective* bit rate of this OFDM system (while accounting for CP overhead)?
- (g) **(ECSE 6560 only)** Given $f_c = 2.4$ GHz, what is the *maximum mobile speed* v_{max} such that ICI remains negligible under the criterion $f_D \leq 0.1 \Delta f$?

OFDM Packet Detection: Sliding Window



- Two windows of L samples each.

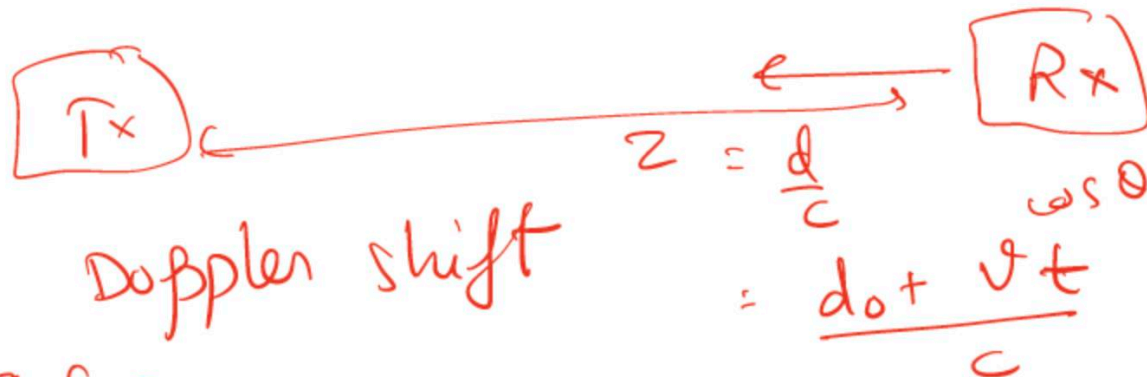
- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

How often we send preamble over time?

Ans: Channel coherence time

mobility

velocity $v \cos \theta$



$$2\pi f_c z =$$

$$2\pi f_c \left(\frac{d_0 + vt}{c} \right) = 2\pi f_c \frac{d_0}{c} + 2\pi f_c \frac{vt}{c}$$

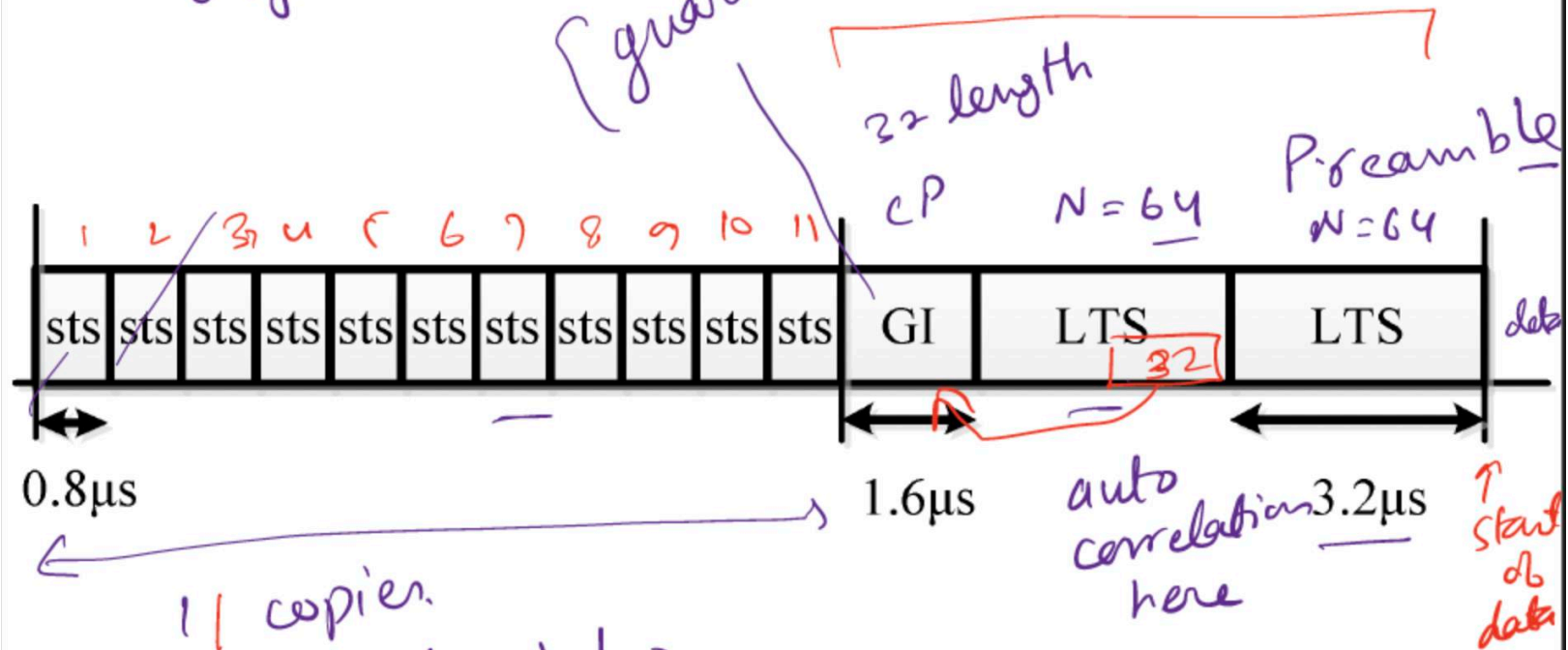
$$f_d = \frac{f_c v \cos \theta}{c} \quad \text{Doppler freq}$$

WiFi Example: STS and LTS

wifi 802.11n

[guard interval]

2.5 copies



0.8 μs

1.6 μs

3.2 μs

11 copies

⇒ sliding window
packet detection
⇒ AGC = Automatic
gain control

auto
correlation
here

↑ start
of data

$f_s = 20 \text{ MHz}$

$T_s = \frac{1}{f_s} = 0.05 \text{ μs}$

$N T_s = 64 \times 0.05 = 3.2 \text{ μs}$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

• Use Preamble to estimate CFO

$$y_1[n] = x[n] e^{-j2\pi \Delta f_c n T_s}$$

$$y_2[n] = x[n] e^{-j2\pi\Delta f_c(nT_s + NT_s)}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

subcarriers
= # FFT size
= # samples in
one OFDM
symbol

 $= NT_s$
= # samples in
Preamble

- Compute:

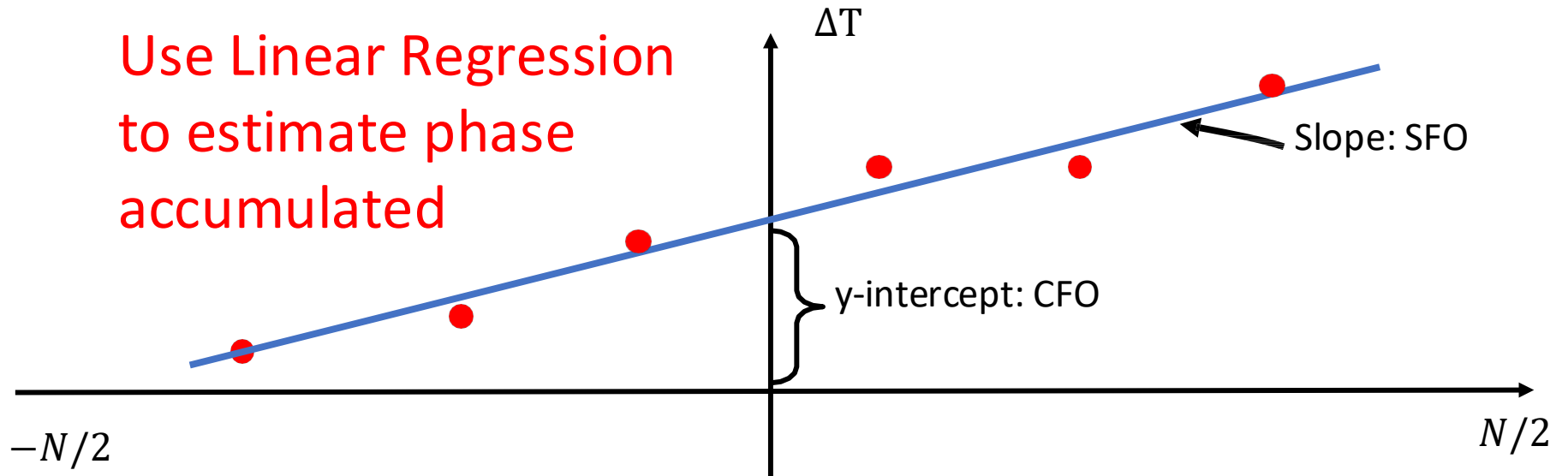
$$A = \sum_{t=1}^N \underbrace{y_1[n]}_{\text{LTS1}} \underbrace{y_2^*[n]}_{\text{LTS2}} = \sum_{t=1}^N \underbrace{x[n]}_{\text{LTS1}} \underbrace{x^*[n]}_{\text{LTS2}} e^{j2\pi\Delta f_c N T_s}$$

same for all n

$$A = e^{j2\pi\Delta f_c N T_s} \sum_{n=1}^N |x[n]|^2 \quad \rightarrow \quad \Delta f_c = \frac{\angle A}{2\pi N T_s}$$

OFDM Phase Tracking

Use Linear Regression
to estimate phase
accumulated



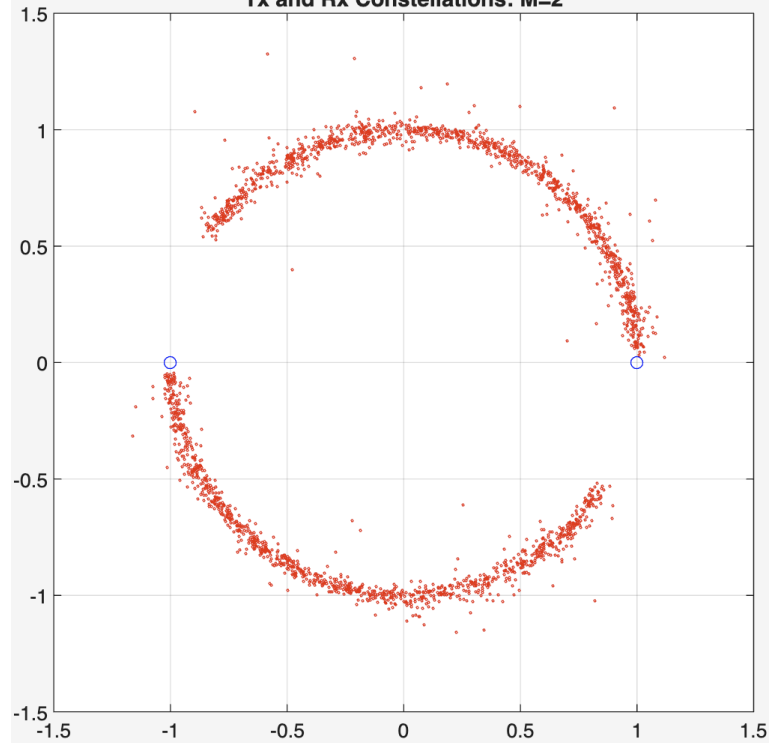
$$\text{Slope} = 2\pi \frac{(N + CP)\delta T_s}{N}$$

$$Y - \text{intercept} = -2\pi\delta f_c(N + CP)T_s$$

$$\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$$

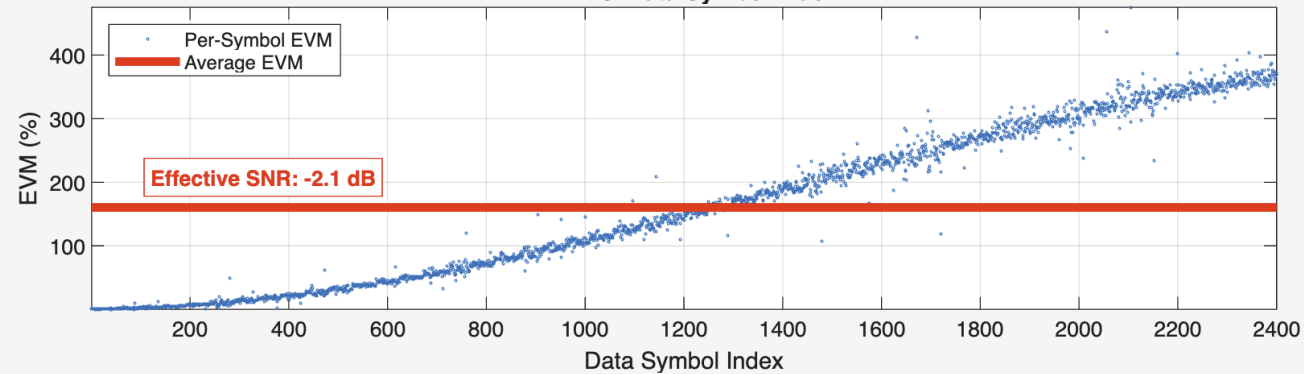
CFO Offset

Tx and Rx Constellations: M=2

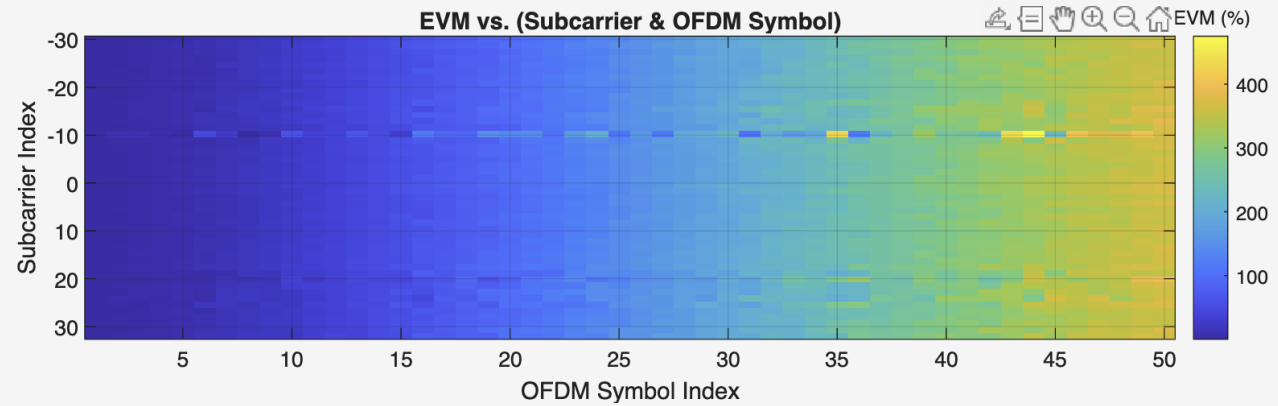


EVM Diagrams

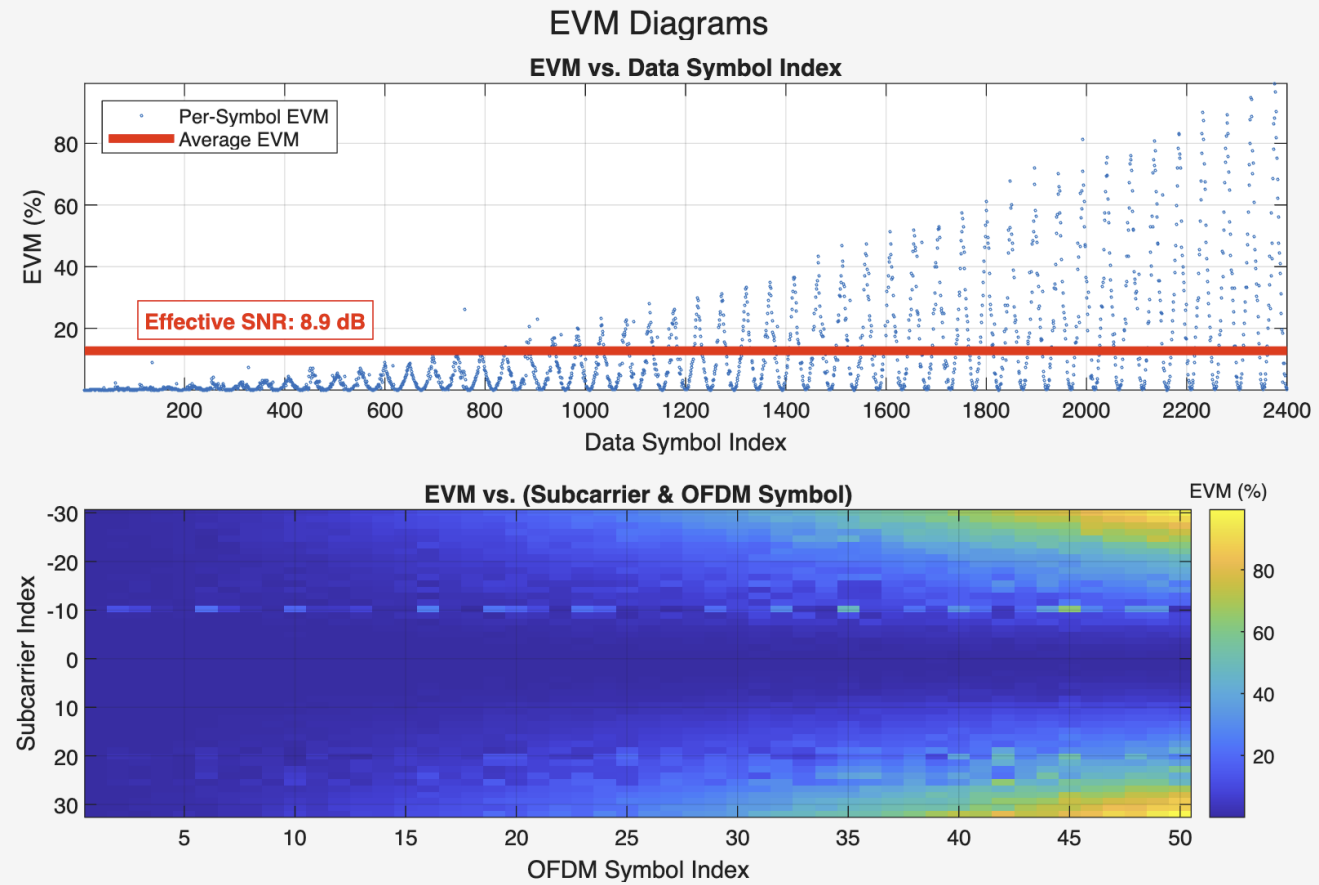
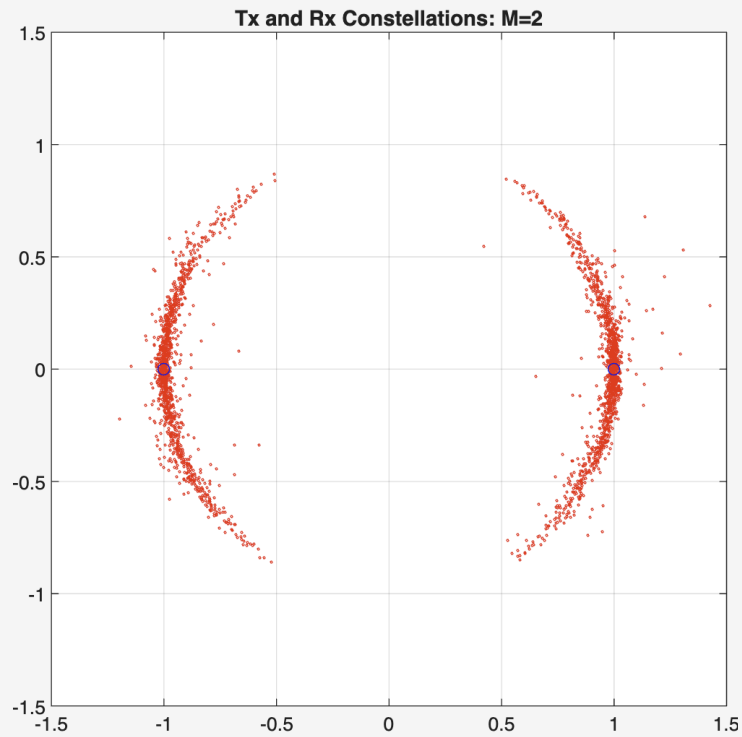
EVM vs. Data Symbol Index



EVM vs. (Subcarrier & OFDM Symbol)



SFO Offset



Question 8 (ECSE 6560 only): Residual CFO and SFO Estimation in OFDM

An OFDM system uses $N = 64$ subcarriers, sampling rate $f_s = 20$ MHz, and cyclic prefix length $N_{CP} = 16$. Two consecutive OFDM data symbols (after coarse synchronization and CFO correction) exhibit the following residual phase offset between their pilots:

$$\Delta\phi(f_k) = 2\pi f_k(N + N_{CP})\delta T_s - 2\pi\delta f_c(N + N_{CP})T_s,$$

where δf_c is the residual carrier frequency offset (CFO) and δT_s is the residual sampling period error (SFO).

- (a) Explain how residual CFO and SFO manifest as a *constant* and a *linear* phase term across subcarriers. Sketch (or describe) the expected pilot phase vs. subcarrier index curve and identify which component corresponds to the CFO and which to the SFO.
- (b) Given the following measured pilot phase differences between two consecutive OFDM symbols:

Subcarrier index (k)	$\Delta\phi_k$ [radians]
-26	+1.90
-13	+1.40
0	+0.90
+13	+0.40
+26	-0.10

Use linear regression $\Delta\phi_k = af_k + b$ to estimate the slope (a) and intercept (b). Take the subcarrier spacing $\Delta f = f_s/N = 312.5$ kHz.

- (c) Compute the residual SFO δT_s and residual CFO δf_c in absolute Hz using:

$$a = 2\pi(N + N_{CP})\delta T_s, \quad b = -2\pi\delta f_c(N + N_{CP})T_s,$$

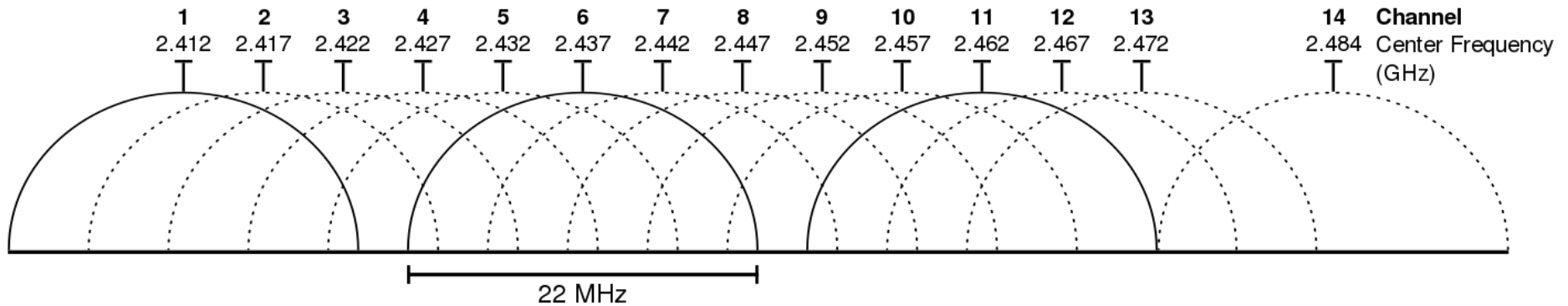
where $T_s = 1/f_s$. Express δT_s as a fractional error (ppm).

- (d) Intuitively describe how residual CFO and SFO affect the constellation across OFDM symbols:
- What visible distortion would residual CFO cause on a constellation diagram?
 - How would residual SFO appear over time if left uncorrected?

Today's Class

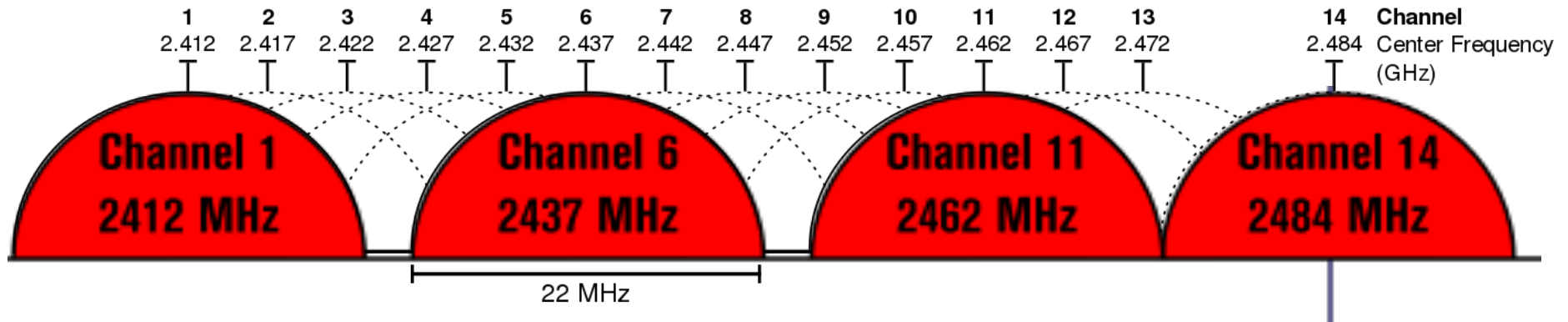
- OFDM Revision
 - HW/Lab 3, 4
 - SFO and Residual Phase correction (ECSE6560 topic)
- **MIMO Intro**
 - **MIMO in WiFi Standards**
 - **MIMO Intro**
 - **MIMO vector representation**

WiFi Standards from WiFi 1 to WiFi 6



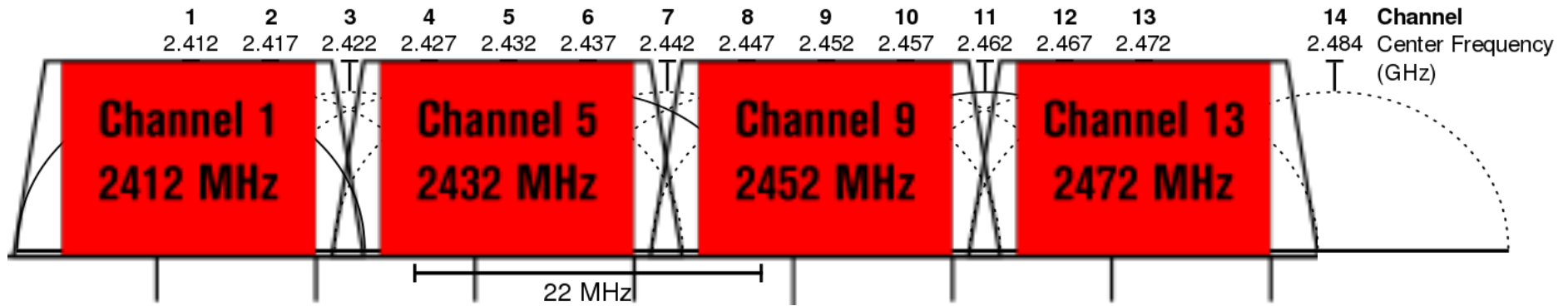
Version	Year	Technology	Modulation	Freq.	Bandwidth	Maximum Data Rate
WiFi 1 (802.11b)	1999	DSSS	DBPSK, DQPSK,	2.4 GHz	22 MHz	11 Mb/s

WiFi Standards from WiFi 1 to WiFi 6



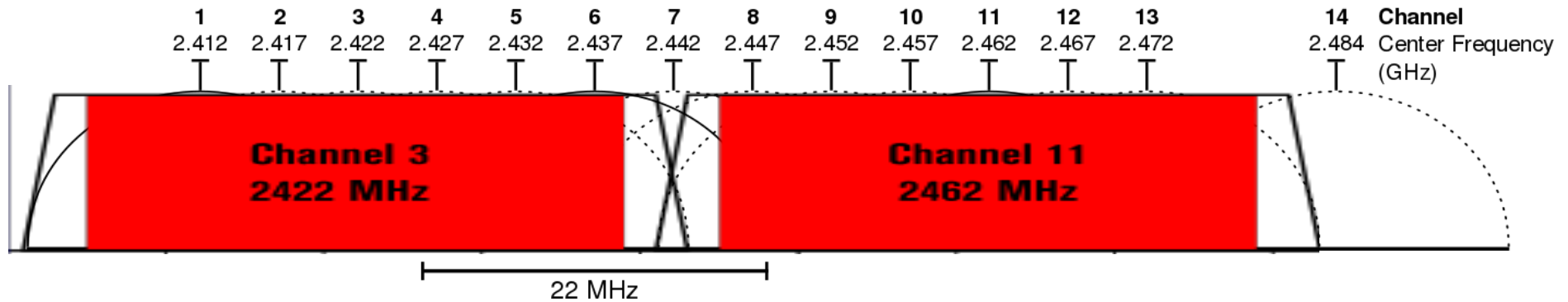
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WiFi Standards from WiFi 1 to WiFi 6



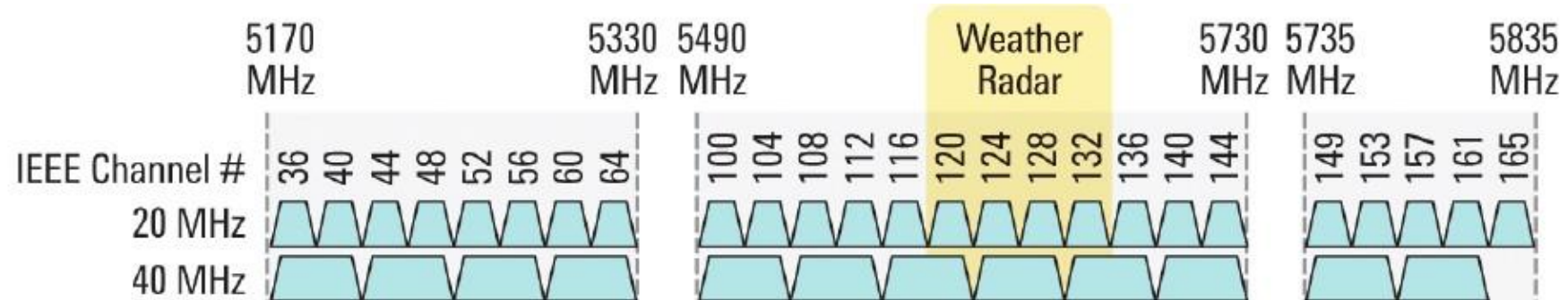
Version	Year	Technology	Modulation	Freq.	Bandwidth	Maximum Data Rate
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WiFi 3 (802.11g)	2003	OFDM (N=64)	BPSK, QPSK, 16-QAM, 64-QAM	2.4 GHz	20 MHz	54 Mb/s

WiFi Standards from WiFi 1 to WiFi 6



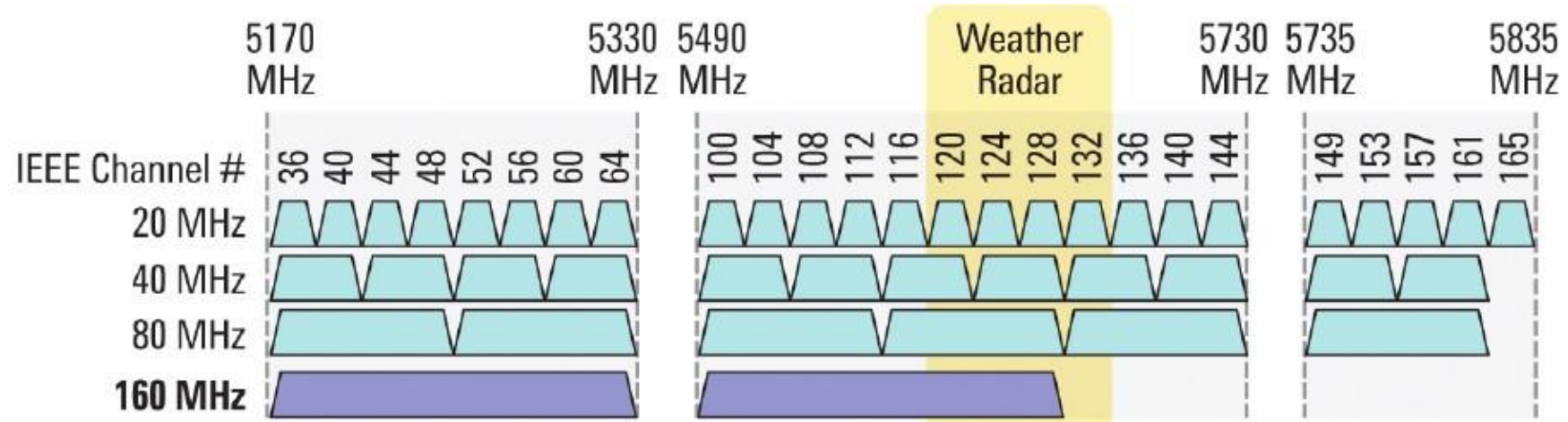
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WiFi 4 (802.11n)	2009	OFDM (N=64) MIMO (4x4)	BPSK, QPSK, 16-QAM, 64-QAM	2.4 GHz 5 GHz	20 MHz 40 MHz	600 Mb/s

WiFi Standards from WiFi 1 to WiFi 6



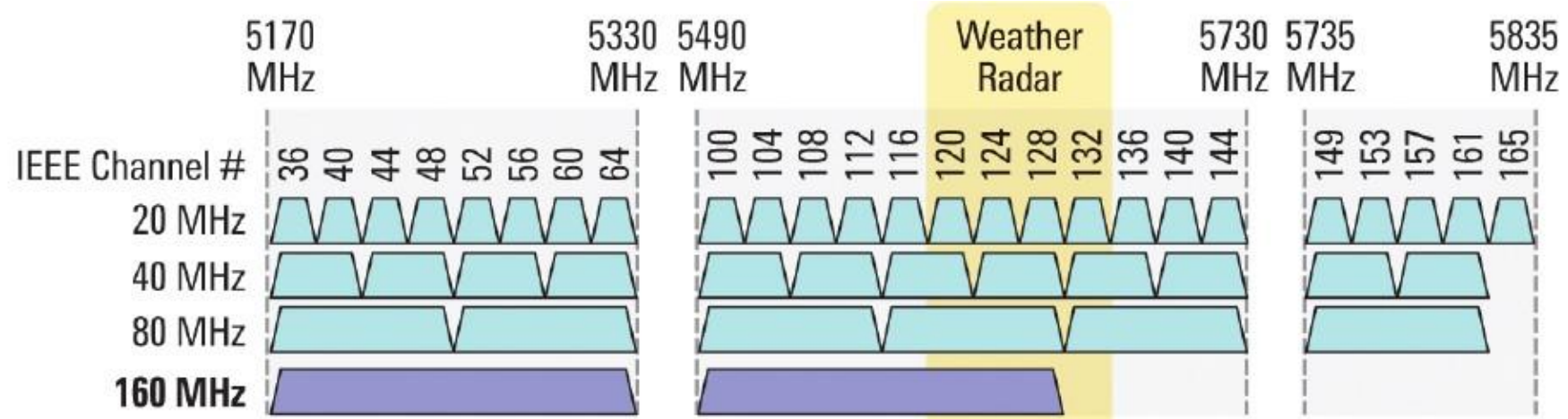
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WiFi Standards from WiFi 1 to WiFi 6



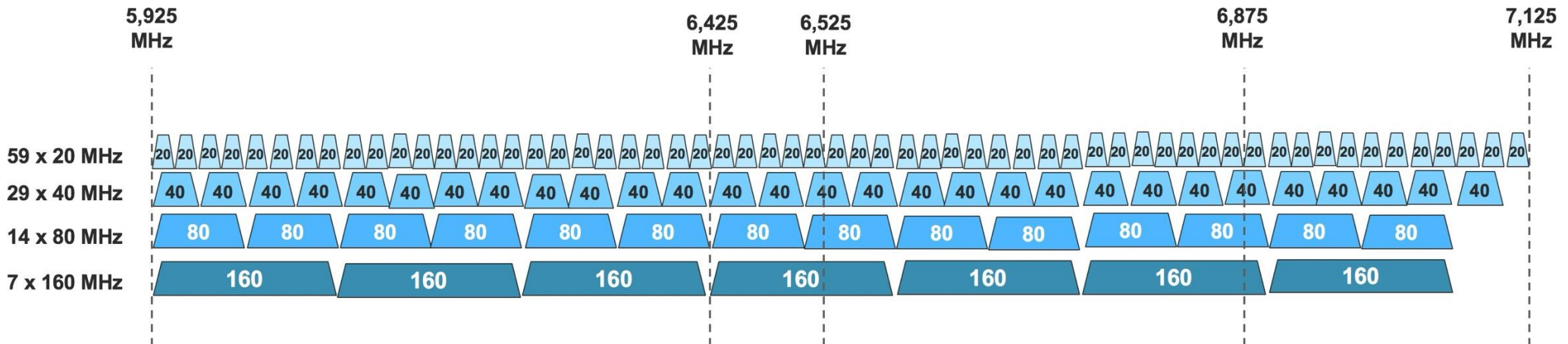
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WiFi 5 (802.11ac)	2014	OFDM (N=64, 128, 256, 512) MIMO (8x8) MU-MIMO (Downlink)	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM	5 GHz	20 MHz 40 MHz 80 MHz 160 MHz	6.933 Gb/s

WiFi Standards from WiFi 1 to WiFi 6



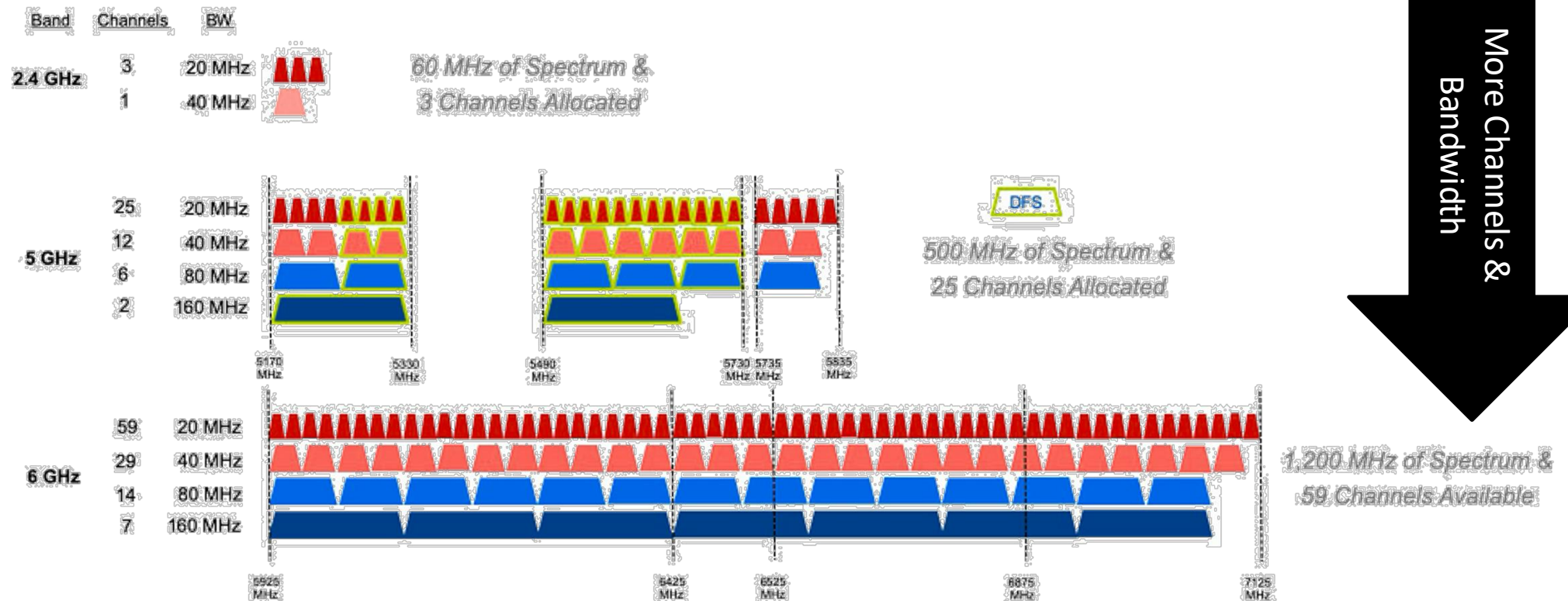
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WiFi 5 (802.11ac)	2014	OFDM (N=64, 128, 256, 512) MIMO (8x8) MU-MIMO (Downlink)	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM	5 GHz	20 MHz 40 MHz 80 MHz 160 MHz	6.933 Gb/s
WiFi 6 (802.11ax)	2019	OFDM (N=256, 512, 1024, 2048) MIMO (8x8) MU-MIMO (Up & Down) OFDMA	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM, 1024-QAM	2.4 GHz 5 GHz	20 MHz 40 MHz 80 MHz 160 MHz	9.608 Gb/s

WiFi Standards from WiFi 1 to WiFi 6



Version	Year	Technology	Modulation	Freq.	Bandwidth	Maximum Data Rate
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WiFi 6 (802.11ax) WiFi 6E	2020	OFDM (N=256, 512, 1024, 2048) MIMO (8x8) MU-MIMO (Up & Down) OFDMA	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM, 1024-QAM	2.4 GHz 5 GHz 6 GHz	20 MHz 40 MHz 80 MHz 160 MHz	9.608 Gb/s

WiFi Standards from WiFi 1 to WiFi 6



WiFi 1

More MIMO Antennas

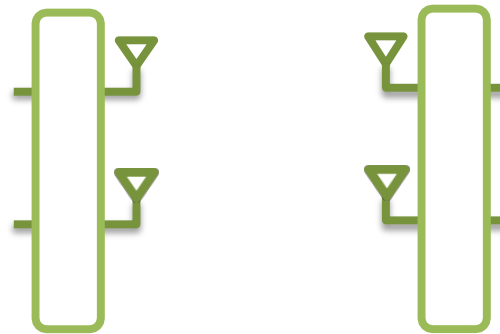
WiFi 5/6

MIMO: Multiple Input Multiple Output

So far: single input single output



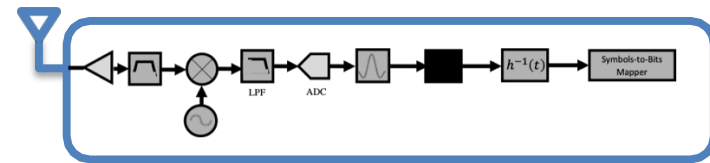
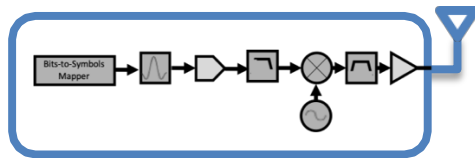
MIMO: multiple input multiple output



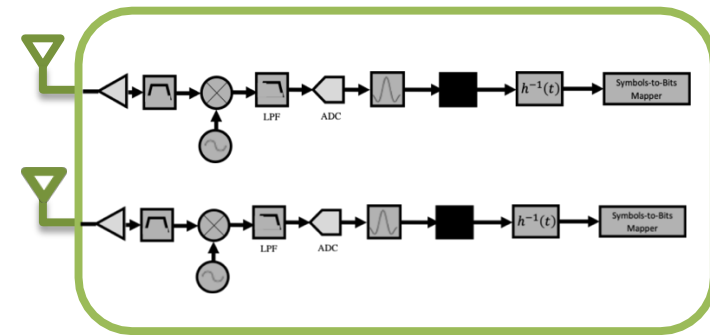
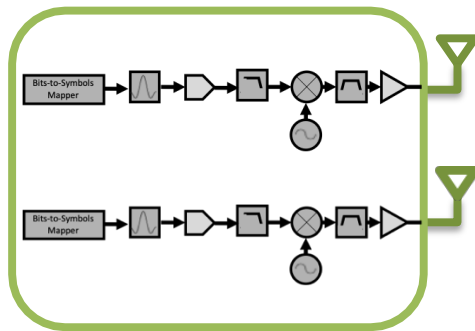
Increase capacity of channel using multiple transmit and receive antennas.

MIMO: Multiple Input Multiple Output

So far: single input single output

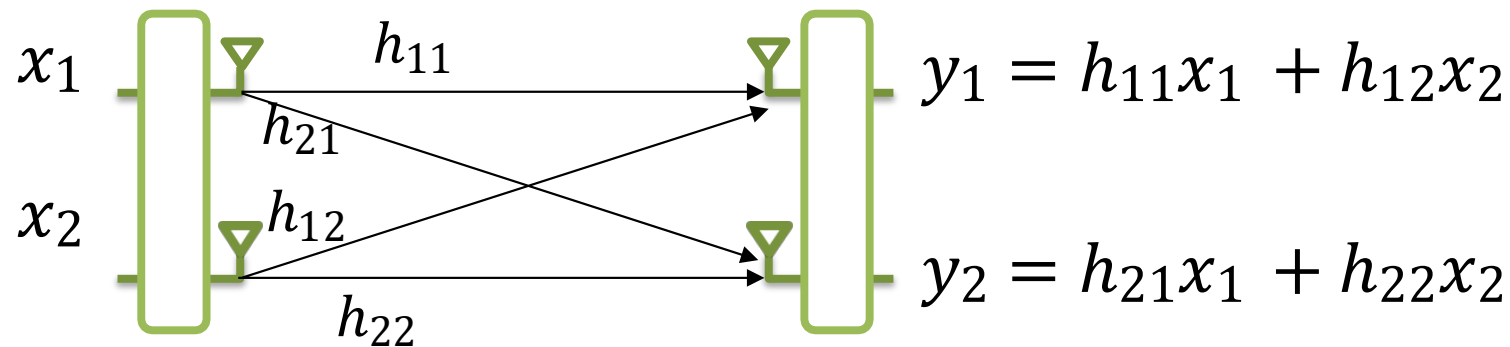


MIMO: multiple input multiple output



Increase capacity of channel using multiple transmit and receive antennas.

MIMO: Multiple TX-RX streams



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

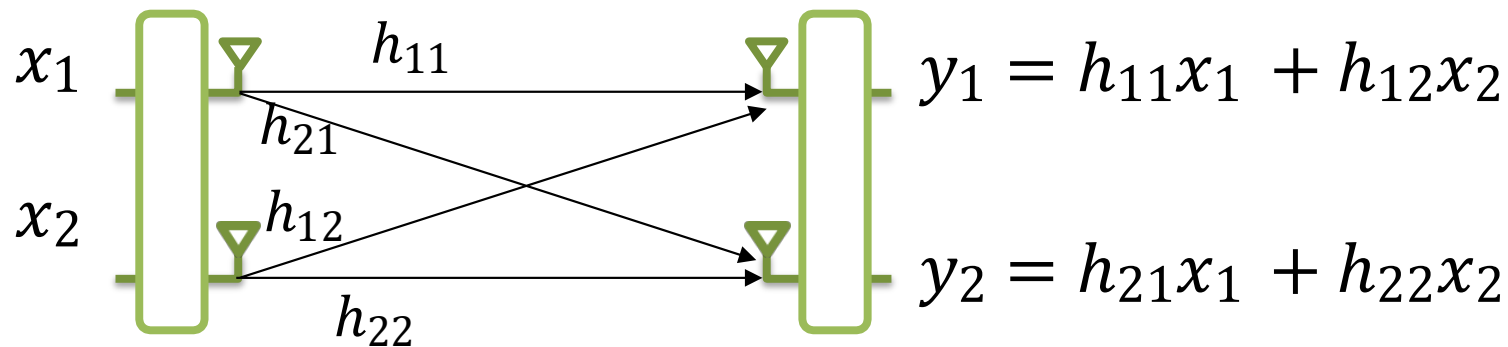
How to recover x_1 and x_2 ?

Estimate $\mathbf{H}$, compute $\mathbf{H}^{-1}$ and invert the channel!

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Transmit 2 packets at the same time!

MIMO: Multiple TX-RX streams



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

How to recover x_1 and x_2 ?

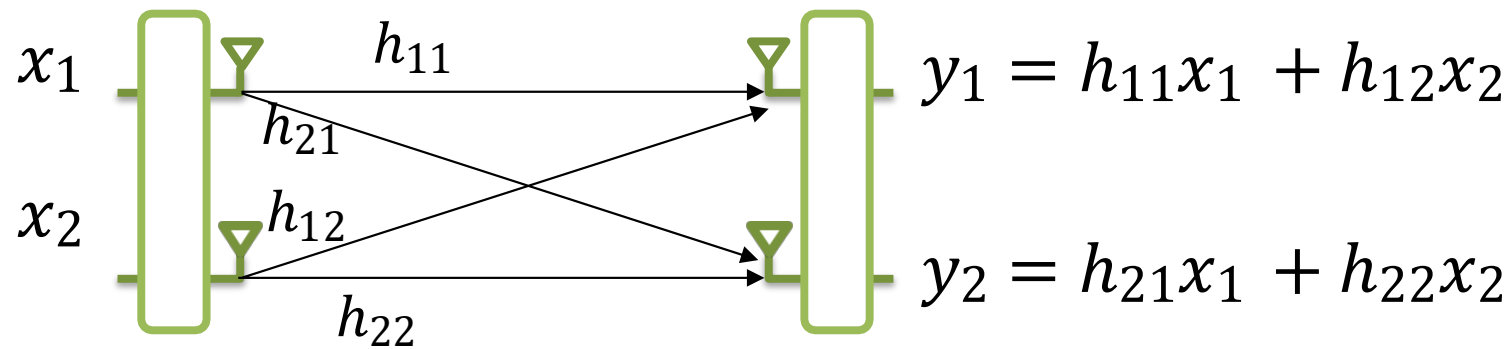
For N antennas, $\mathbf{H}$ is $N \times N$ matrix $\rightarrow O(N^3)$

Estimate $\mathbf{H}$, compute $\mathbf{H}^{-1}$ and invert the channel!

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Noise amplification

MIMO: Vector Representation



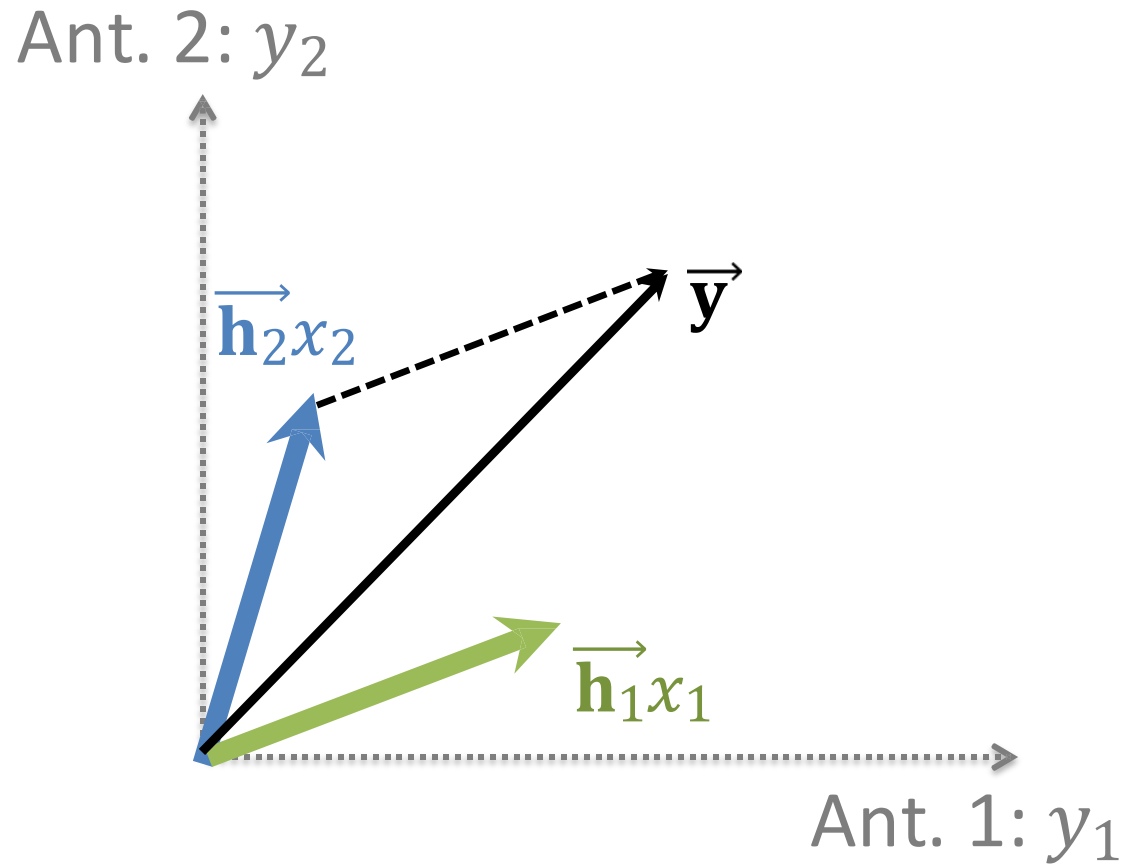
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

How to recover x_1 and x_2 ?

$$\mathbf{y} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2 = \overrightarrow{\mathbf{h}}_1 x_1 + \overrightarrow{\mathbf{h}}_2 x_2$$

MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$



MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$

To decode x_1 , project on a vector $\vec{h}_2^\perp$ orthogonal to $\vec{h}_2$

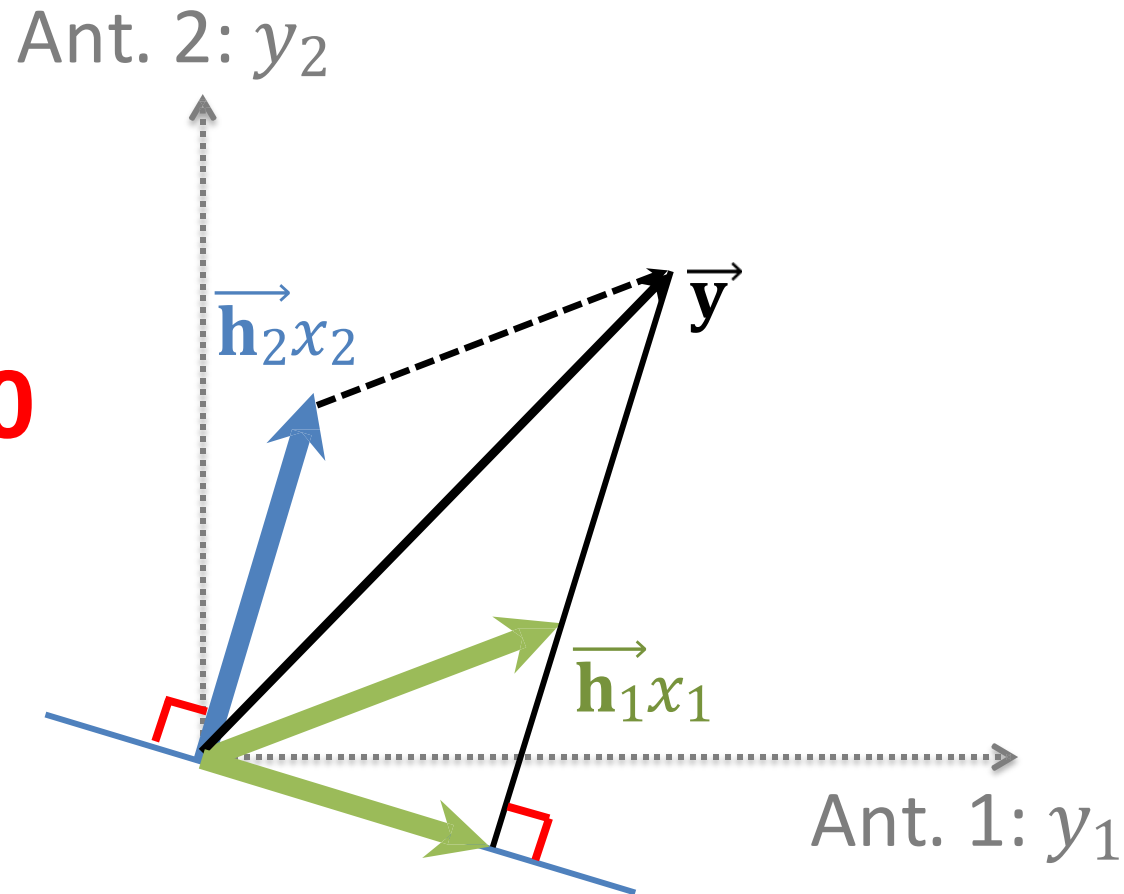
$$\vec{h}_2^\perp \vec{y} = \vec{h}_2^\perp \vec{h}_1 x_1 + \vec{h}_2^\perp \vec{h}_2 x_2$$

$$= \vec{h}_2^\perp \vec{h}_1 x_1$$

$$\mathbf{y} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2$$

$$\vec{h}_2^\perp = \begin{bmatrix} h_{22} & -h_{12} \end{bmatrix}$$

$$\begin{aligned} \vec{h}_2^\perp \vec{y} &= h_{22} h_{11} x_1 - h_{12} h_{21} x_1 \\ &\quad + \cancel{h_{22} h_{12} x_2} - \cancel{h_{12} h_{22} x_2} \\ &= (h_{22} h_{11} - h_{12} h_{21}) x_1 \end{aligned}$$



MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$

To decode x_2 , project on a vector $\vec{h}_1^\perp$ orthogonal to $\vec{h}_1$

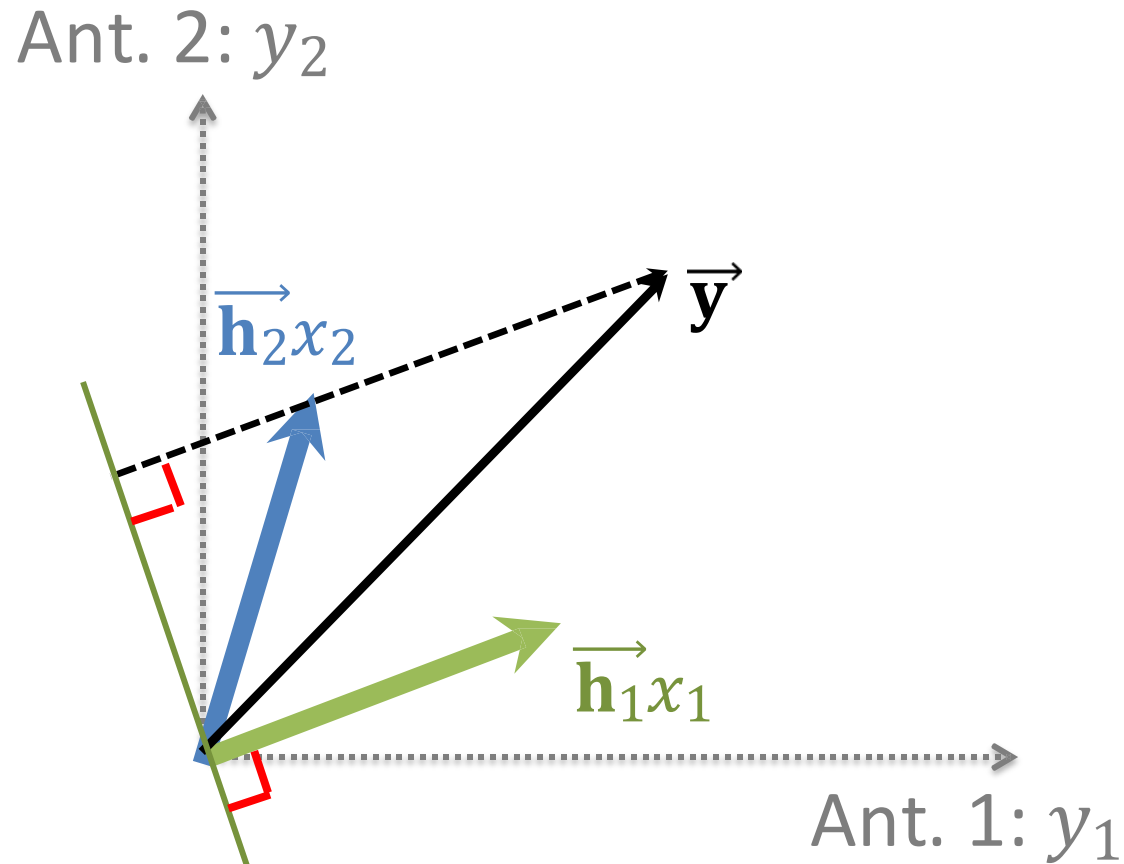
$$\vec{h}_1^\perp \vec{y} = \vec{h}_1^\perp \vec{h}_1 x_1 + \vec{h}_1^\perp \vec{h}_2 x_2$$

$$= \vec{h}_1^\perp \vec{h}_2 x_2$$

$$\mathbf{y} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2$$

$$\vec{h}_1^\perp = \begin{bmatrix} h_{21} & -h_{11} \end{bmatrix}$$

$$\begin{aligned} \vec{h}_1^\perp \vec{y} &= \cancel{h_{21} h_{11} x_1} - \cancel{h_{21} h_{11} x_1} + h_{21} h_{12} x_2 - h_{11} h_{22} x_2 \quad \mathbf{0} \\ &= (h_{21} h_{12} - h_{11} h_{22}) x_2 \end{aligned}$$

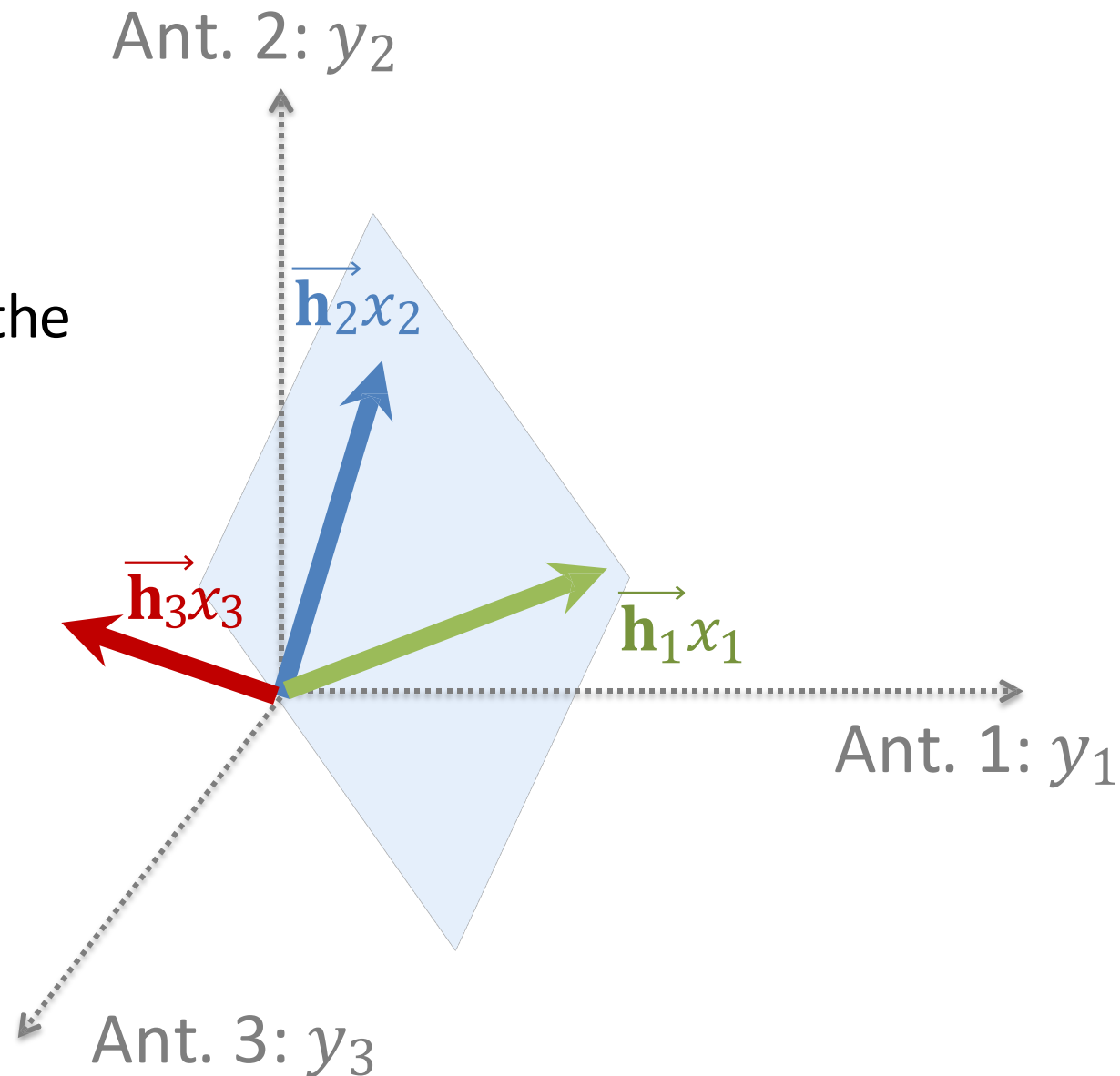


MIMO: Antenna Space

What about MIMO with more antennas?

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2 + \vec{h}_3 x_3$$

To decode x_1 , project on a vector $\vec{h}_{23}^\perp$ orthogonal to the plane formed by $\vec{h}_2$ and $\vec{h}_3$

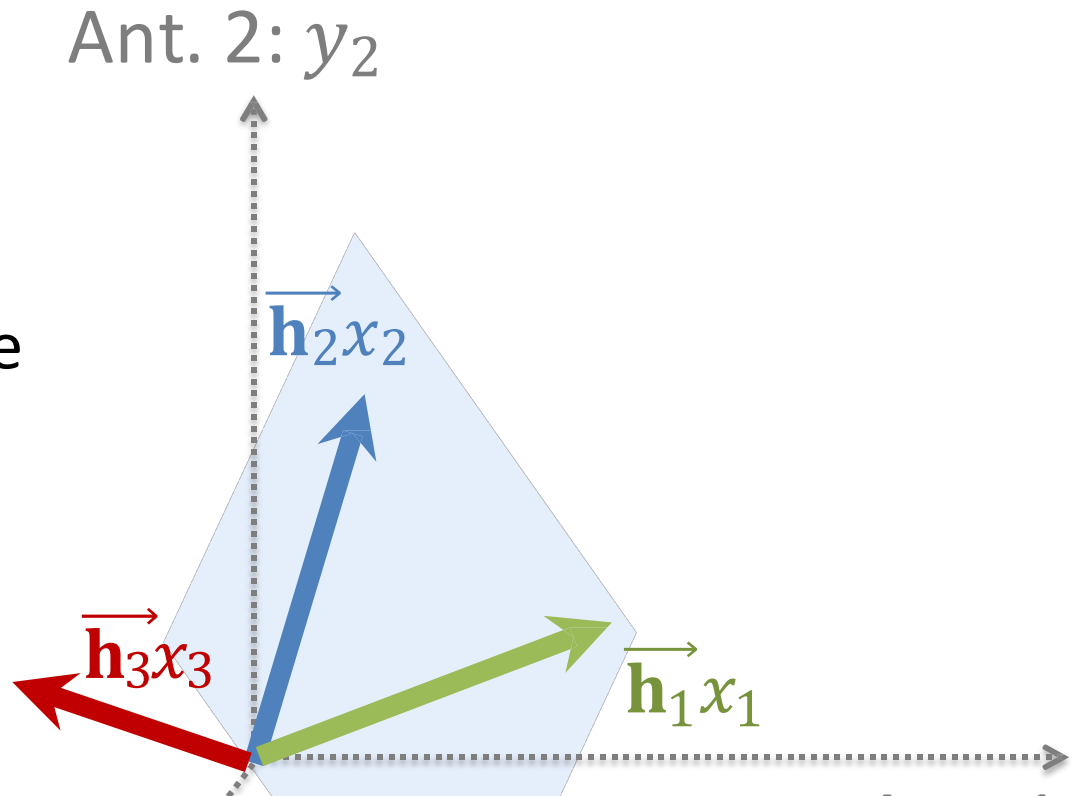


MIMO: Antenna Space

What about MIMO with more antennas?

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2 + \vec{h}_3 x_3$$

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N antenna MIMO, receives signals in N dimensional space
→ Can decode N parallel signals